TIME
2024-04

CMOS電路設計

CMOS Circuit Design

班別: 產業新尖兵 - 科技與AI應用人才培訓班 (第一梯次)

學號: 007

報告人: 王維碩

課堂講師: 陳建至

Designed by Wei Shuo Wang in Kaohsiung Use Canva.

大綱
Outline

01 基本介紹
Introduction

02 數位電路
Digital Circuit

03 類比電路
Analog Circuit

04 總結
Summary

01

基本介紹
Introduction

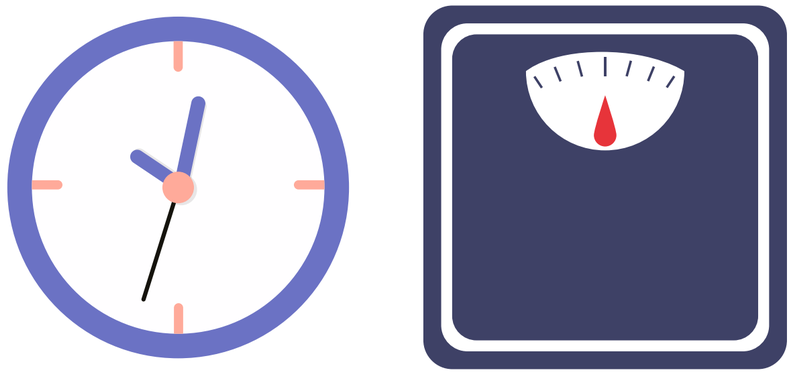
電子電路有哪些呀?

數位 VS. 類比

Digital VS. Analog

- **類比電路 (Analog Circuit)**

  可以顯示出連續訊號、電壓和電流的變化。



疑? 好像變肥了??

- **數位電路 (Digital Circuit)**

  離散信號 (只有0、1)，電壓和電流的狀態H、L表示。

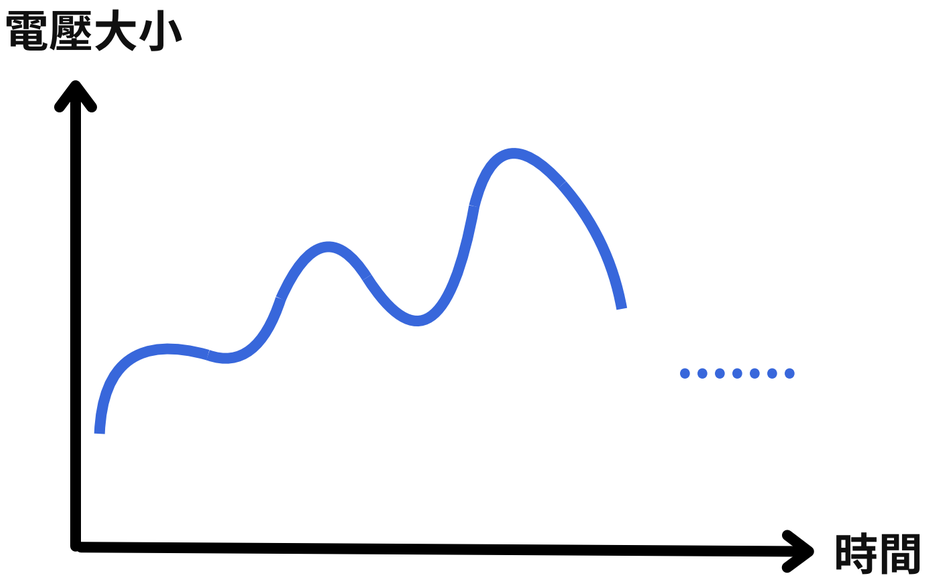
啊~ 絕對變胖了

差異

Difference

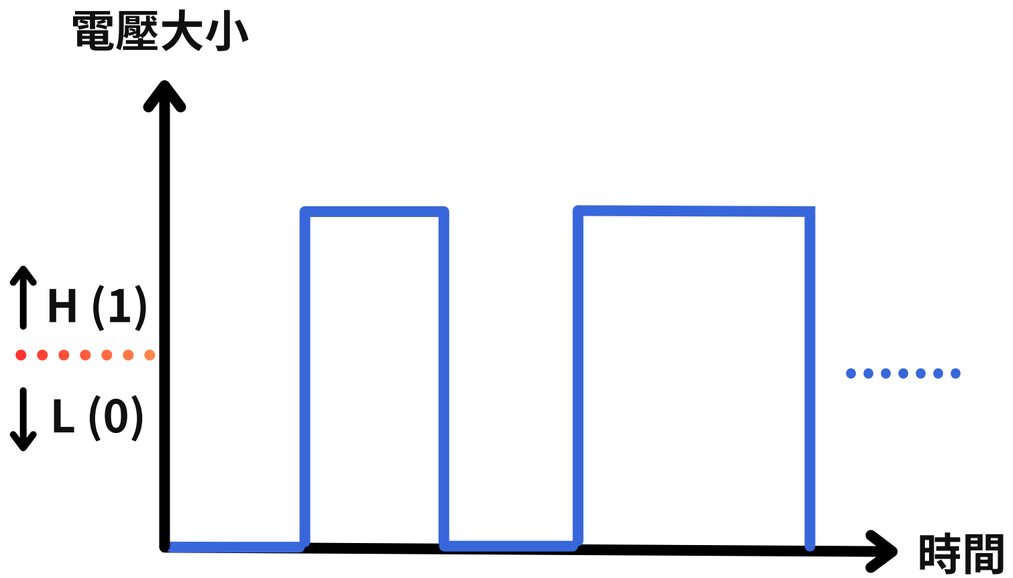
- **類比電路 (Analog Circuit)**

  可以顯示出連續訊號、電壓和電流的變化。

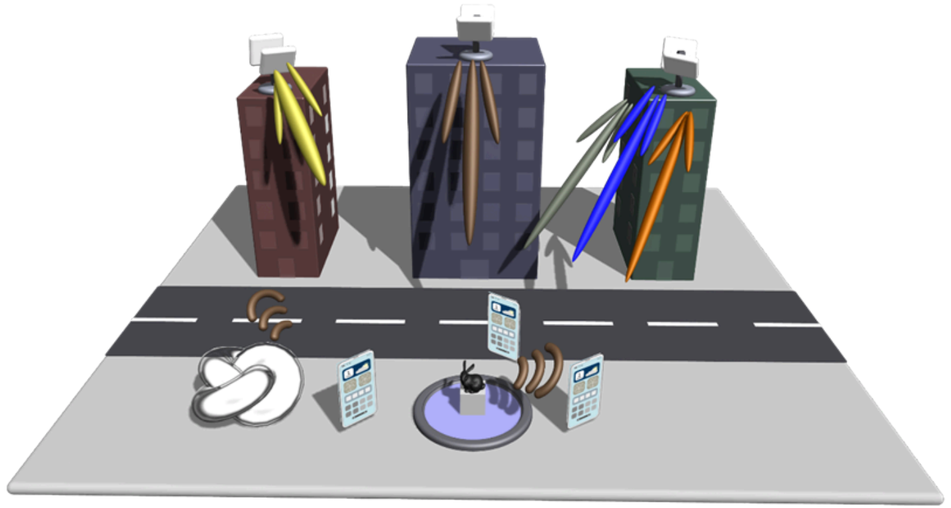


- **數位電路 (Digital Circuit)**

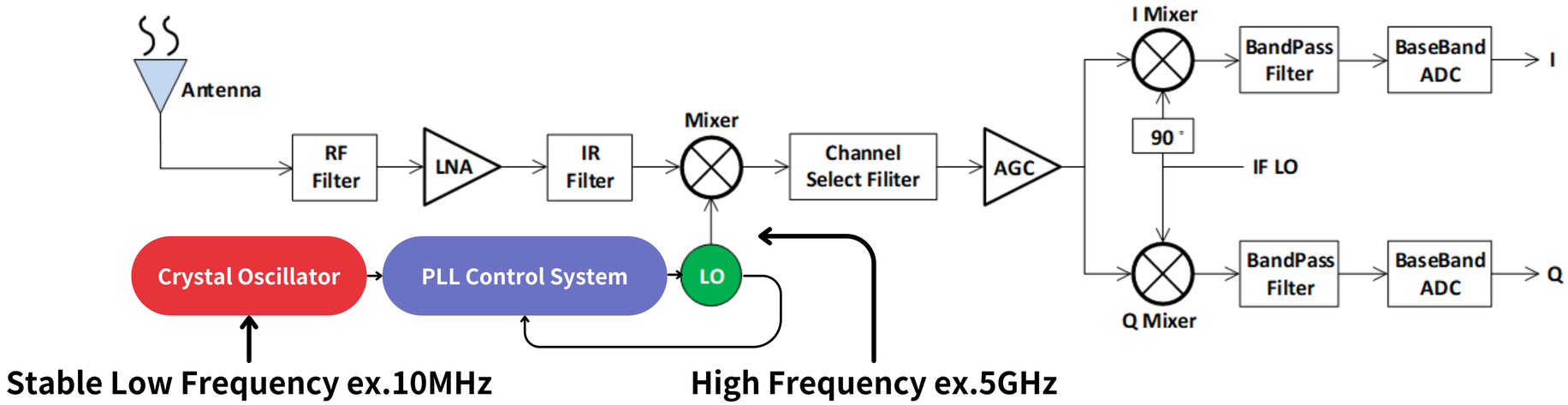
  離散信號 (只有0、1)，電壓和電流的狀態H、L表示。



應用 Application



▲波束賦形天線與裝置移動後自動配對的MIMO系統



▲接收機架構

0 1
not
or
and

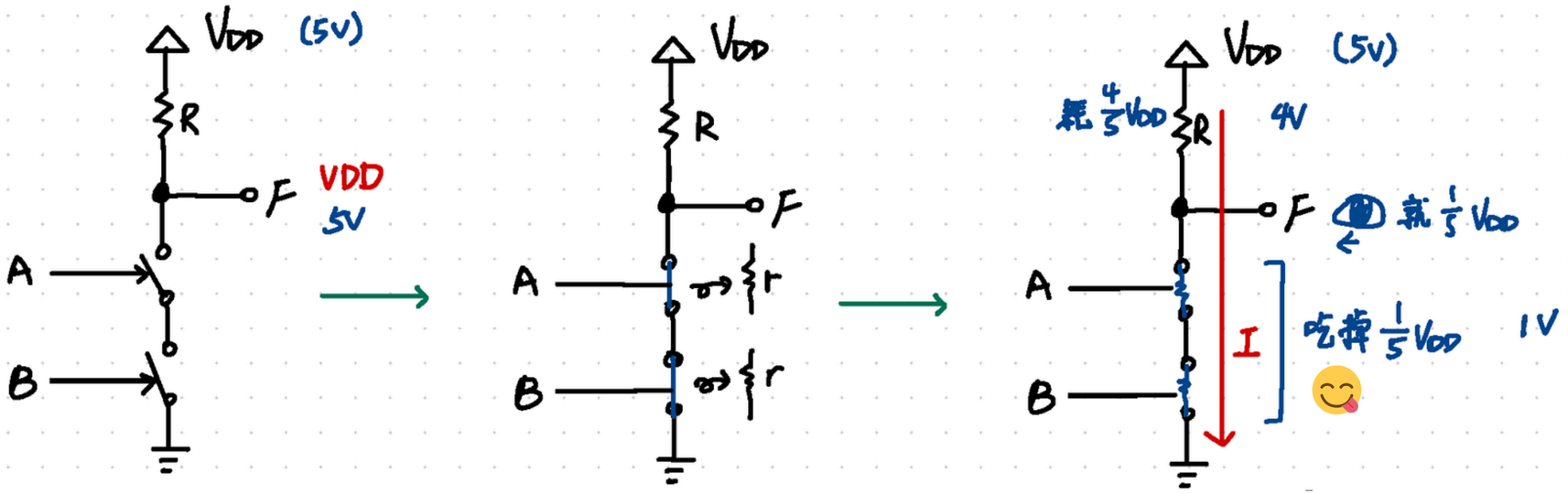
02

數位電路
Digital Circuit

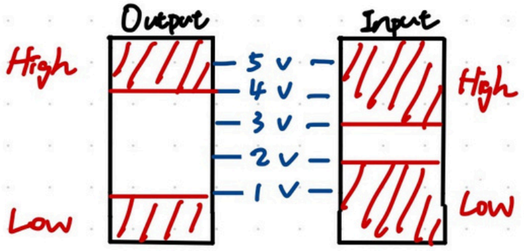
關於數位電路，我的認識的是.....

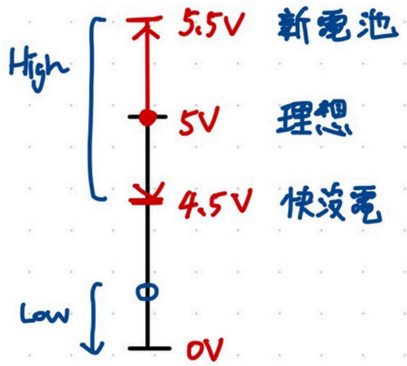
數位電路判斷

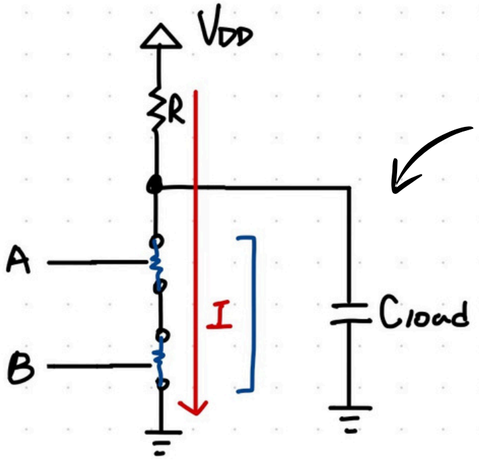
Digital Circuit 0 or 1



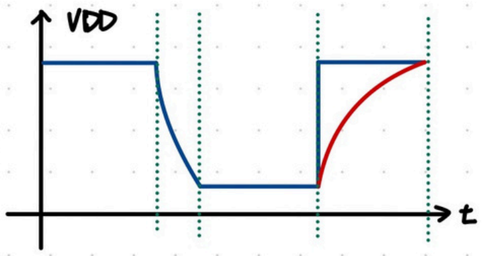
- High、Low 做出範圍
- Input能有更多寬容度
- Input A、B (開): F=5V
- Input A、B (關): F=1V



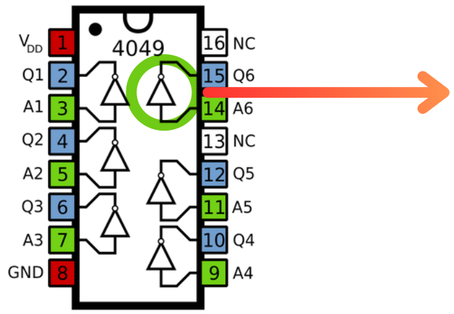




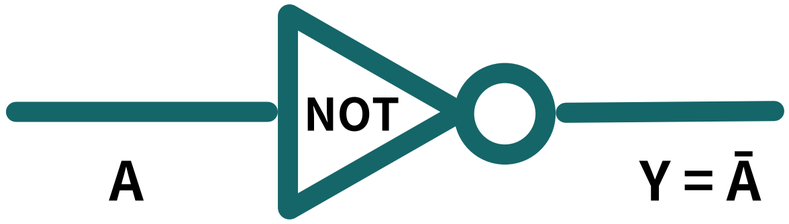
模擬後級電路造成的寄生電容
多餘的電容會使得電路出現非預期效果



數位邏輯閘 (Digital Logic Gates)

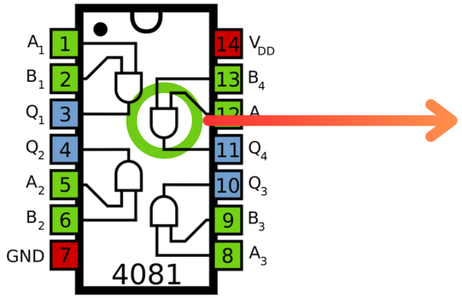


😍 喜歡



😫 討厭

命運的捉弄



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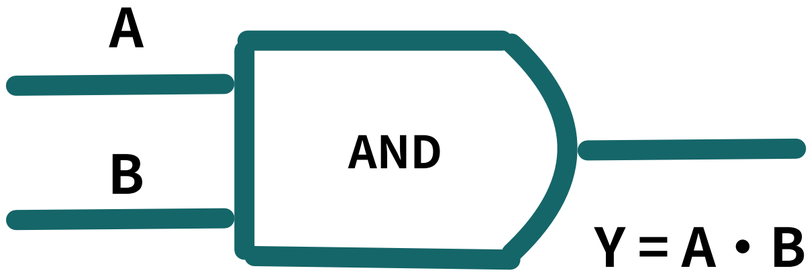
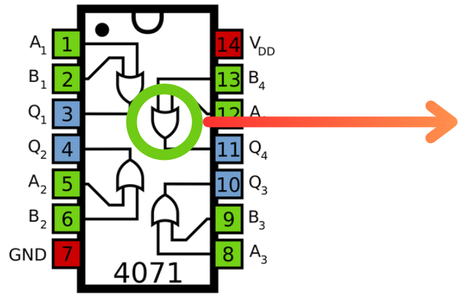


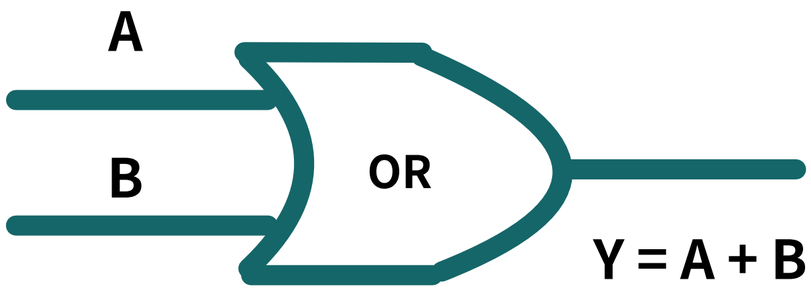
😍 喜歡

完美結果



😍 喜歡

😫 討厭



😍 喜歡

單戀

參考書目: 天野英晴 , “世界第一簡單數位電路” , 世茂出版有限公司,ISBN:9789865779719, 2015.

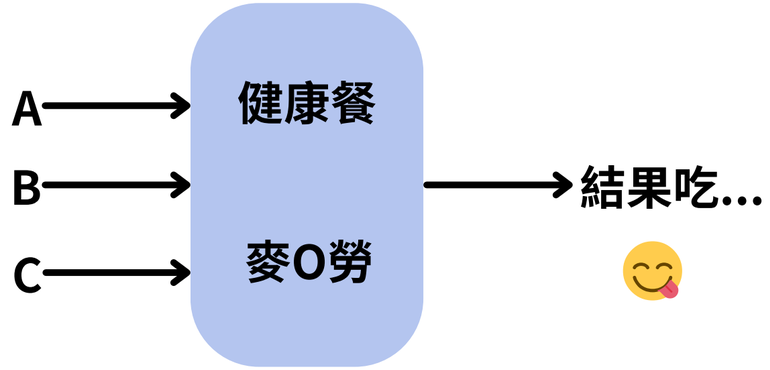
數位電路

Digital Circuit

- **Combinatorial logic (組合邏輯電路)**

→ 由當前的輸入組合，決定輸出的電路。

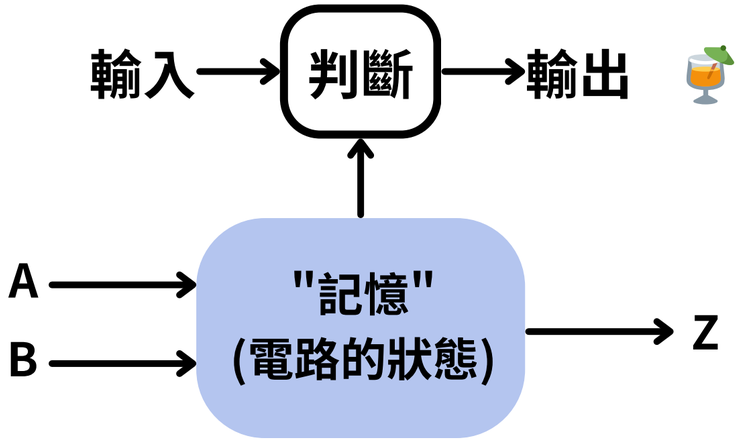
EX: 健康餐、麥O勞，用來投票決定吃什麼的電路。



- **Sequential logic（序向邏輯電路)**

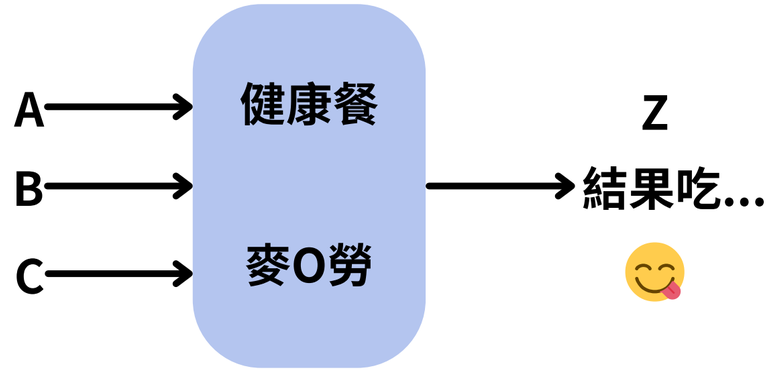
→ 由當前的輸入組合和"記憶"(先前的輸入)，決定輸出的電路。

EX: 自動販賣機，先投入5元，再投入10元，機器顯示15元，
就可以挑選該金額可以購買的飲料。



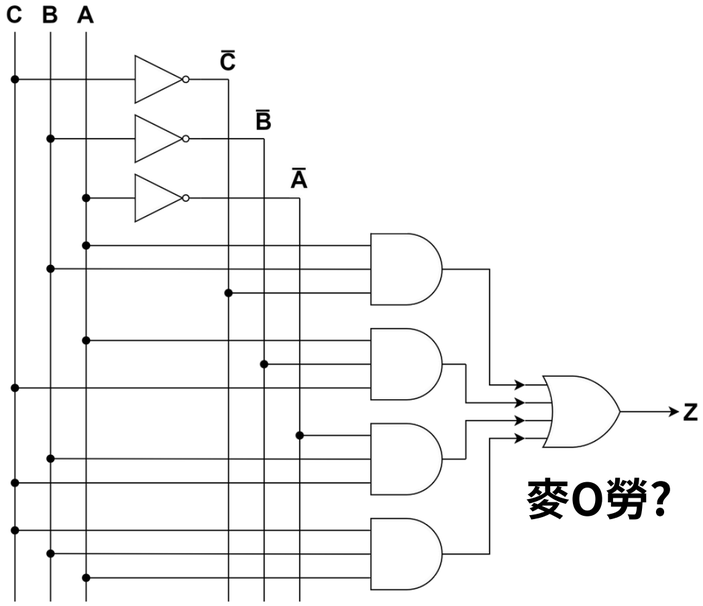
組合邏輯電路

Combinatorial Logic Circuit



C	B	A	Z
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

0:健康餐　1:麥O勞

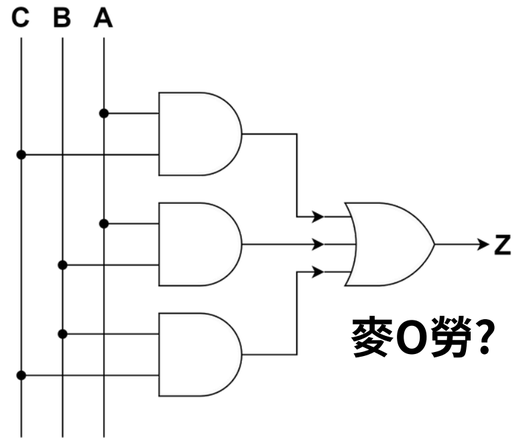


卡諾圖化簡電路

C \ BA	00	01	11	10
0	0	0	1	0
1	0	1	1	1

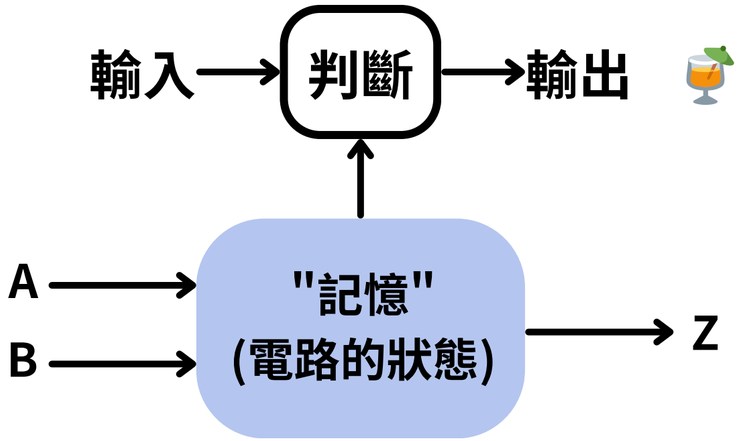
C • A　B • A　C • B

註.
長寬1、2、4長方形或正方形
尋找共通點，群組間重複圈選尤佳
數量越少越好、圈選範圍越大越好

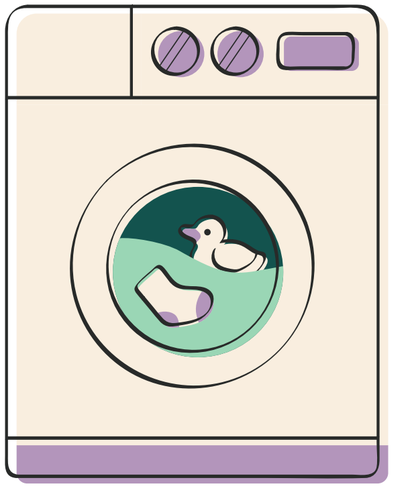


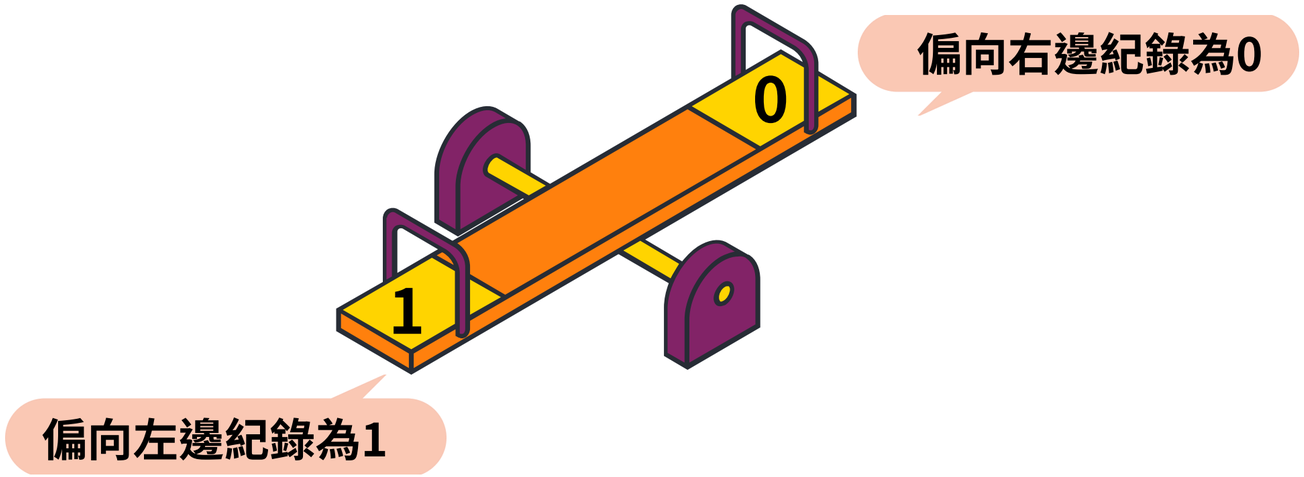
序向邏輯電路

Sequential Logic Circuit



- 正反器 Flip-flop



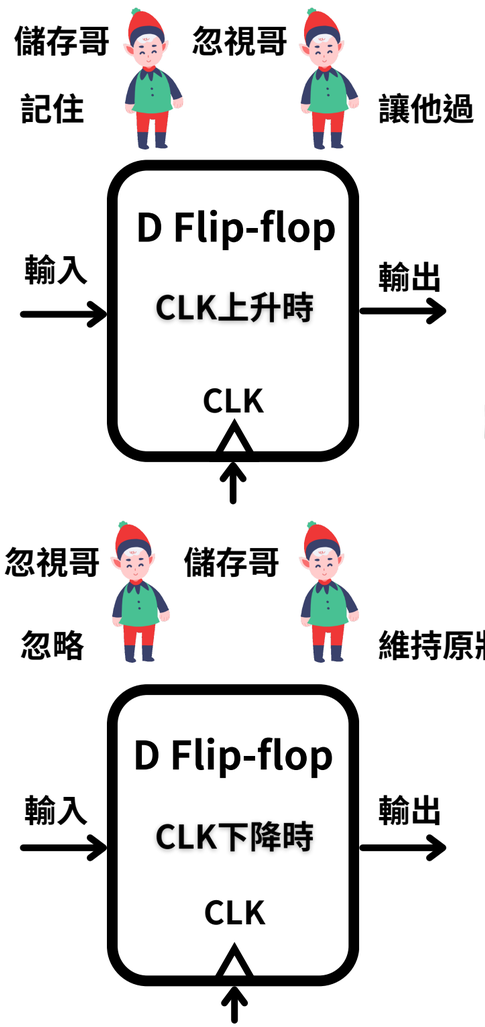


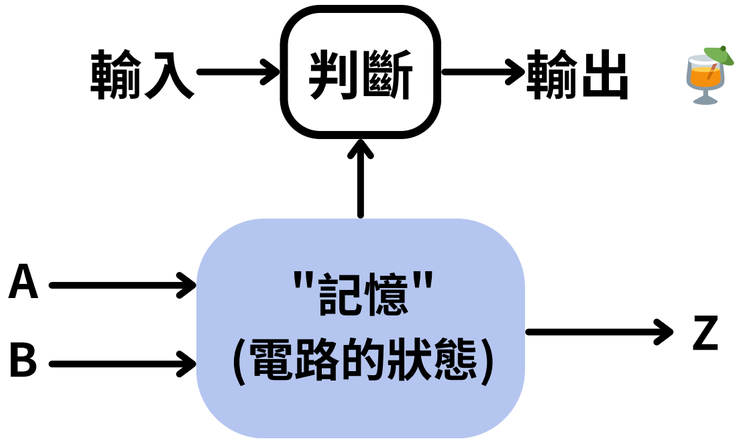
洗衣機翻滾衣物發出啪搭啪搭聲音，可以儲存狀態的元件，可記錄二進位數位訊號「1」和「0」。

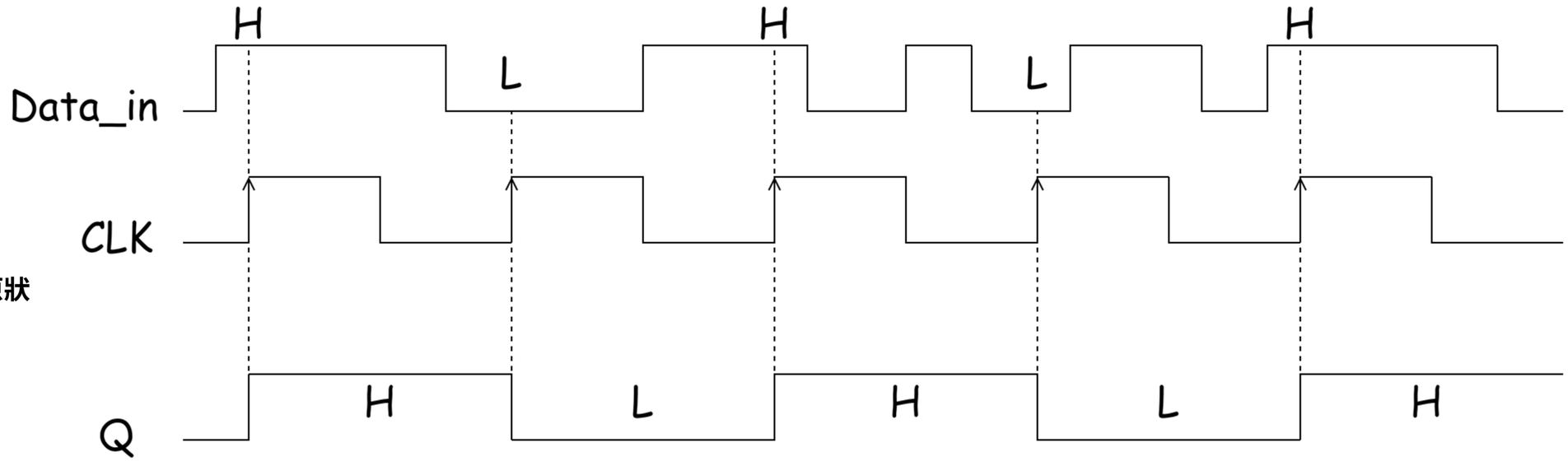
參考書目: 天野英晴，“世界第一簡單數位電路”, 世茂出版有限公司,ISBN:9789865779719, 2015.

序向邏輯電路

Sequential Logic Circuit

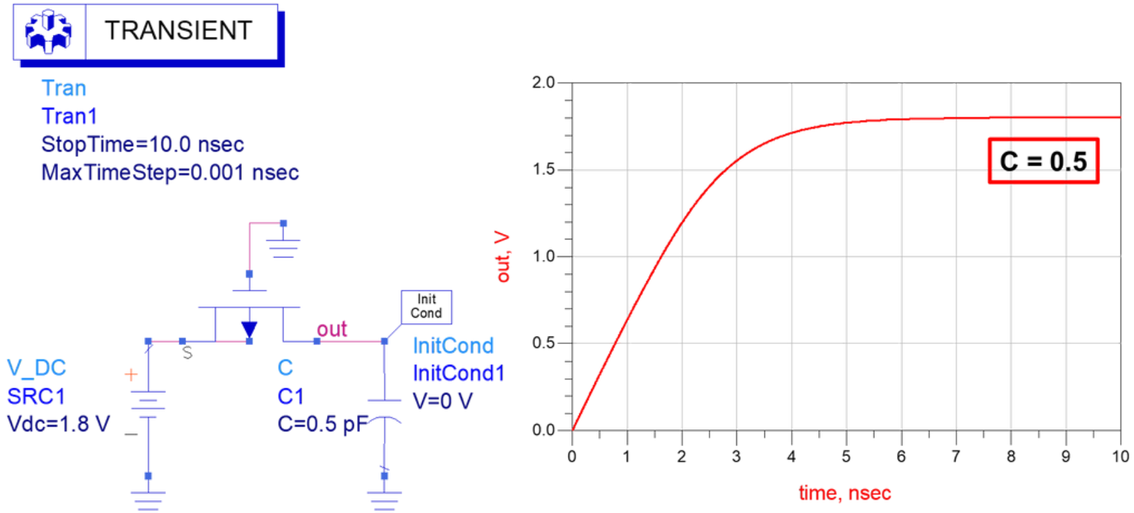


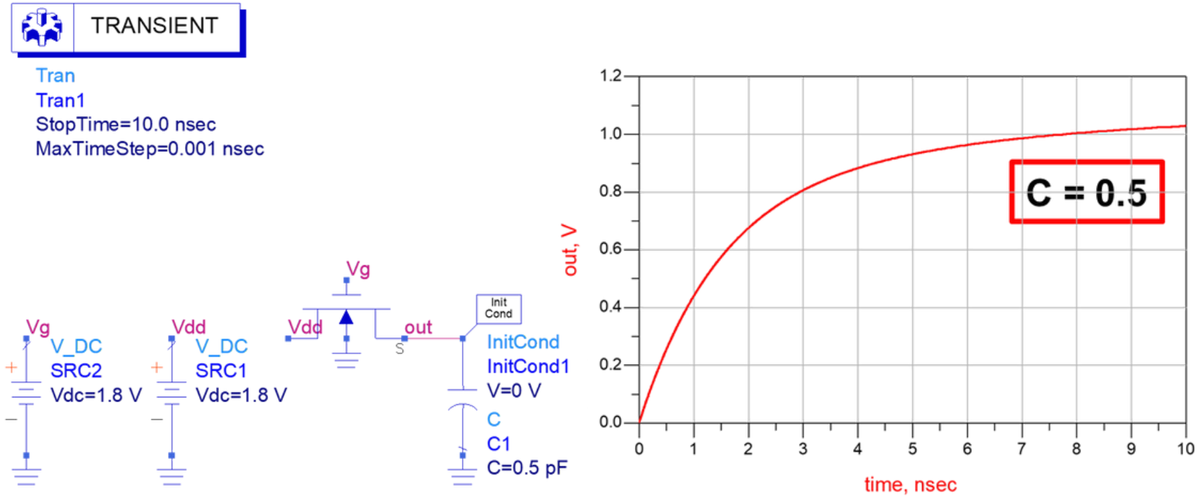


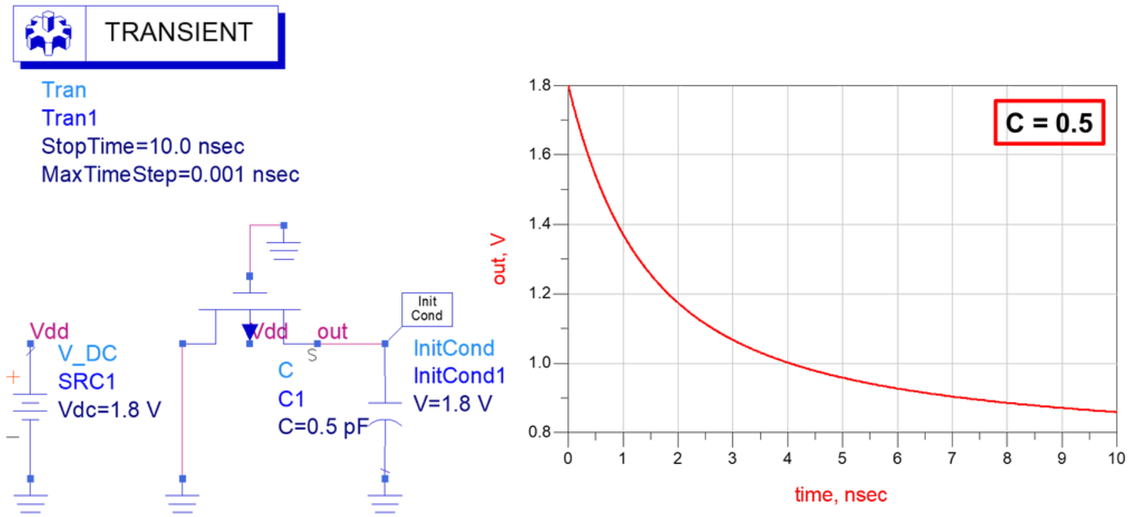


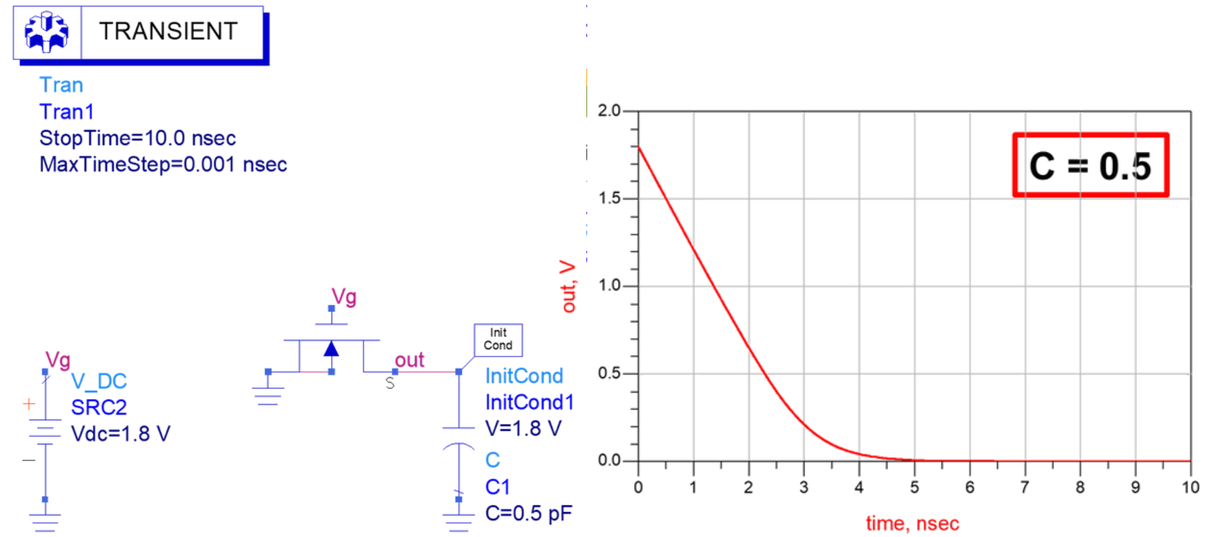
- 正反器時序

為何需要CMOS？



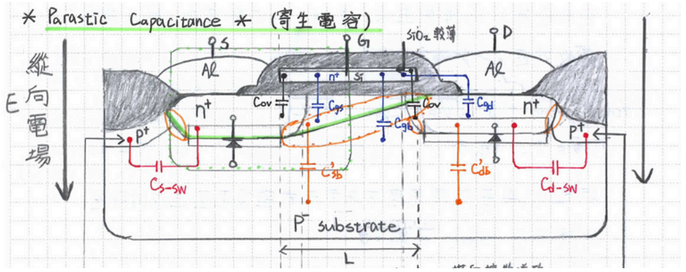


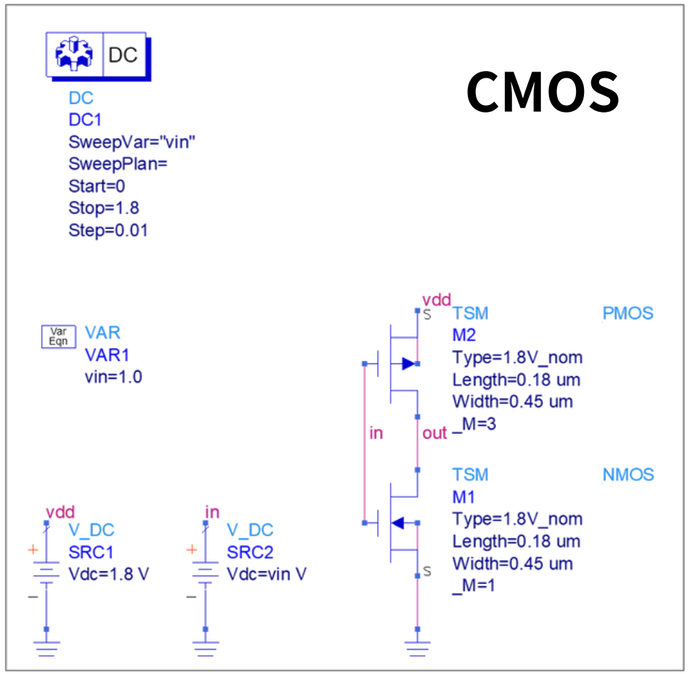


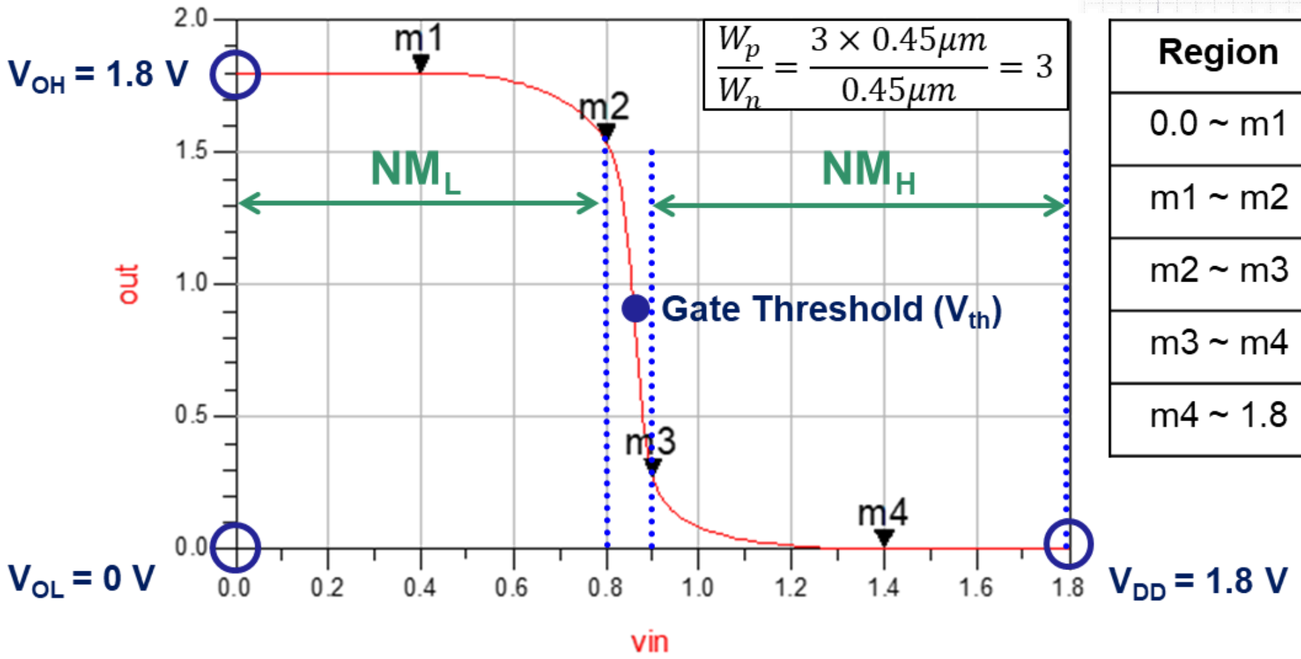


- PMOS 充放電
- NMOS 充放電

為何需要CMOS？







Region	nMOS	pMOS
0.0 ~ m1	off	triode
m1 ~ m2	saturated	triode
m2 ~ m3	saturated	saturated
m3 ~ m4	triode	saturated
m4 ~ 1.8	triode	off

- $\frac{V_n}{V_p} = \frac{\mu_n E}{\mu_p E} \cong \frac{1350^{cm^2/_{v \cdot s}}}{480^{cm^2/_{v \cdot s}}} = 2 \sim 3$倍

- $\frac{i_{DP}}{i_{DN}} = \frac{\frac{1}{2}\mu_p Cox \left(\frac{W}{L}\right)_p (V_{GS}-V_t)^2}{\frac{1}{2}\mu_n Cox \left(\frac{W}{L}\right)_n (V_{GS}-V_t)^2} = 1$, if $\frac{\mu_p}{\mu_n} = \frac{1}{3}$, $\frac{Cox\left(\frac{W}{L}\right)_p}{Cox\left(\frac{W}{L}\right)_n} = \frac{3}{1}$

註. μ: 載子遷移率（carrier mobility）、W: MOSFET的閘極寬度、L: MOSFET的閘極長度 、Cox: 閘極氧化層的單位電容大小

在相同的電壓作用下，如果要產生相同的電流，PMOS之面積要為NMOS的2~3倍。

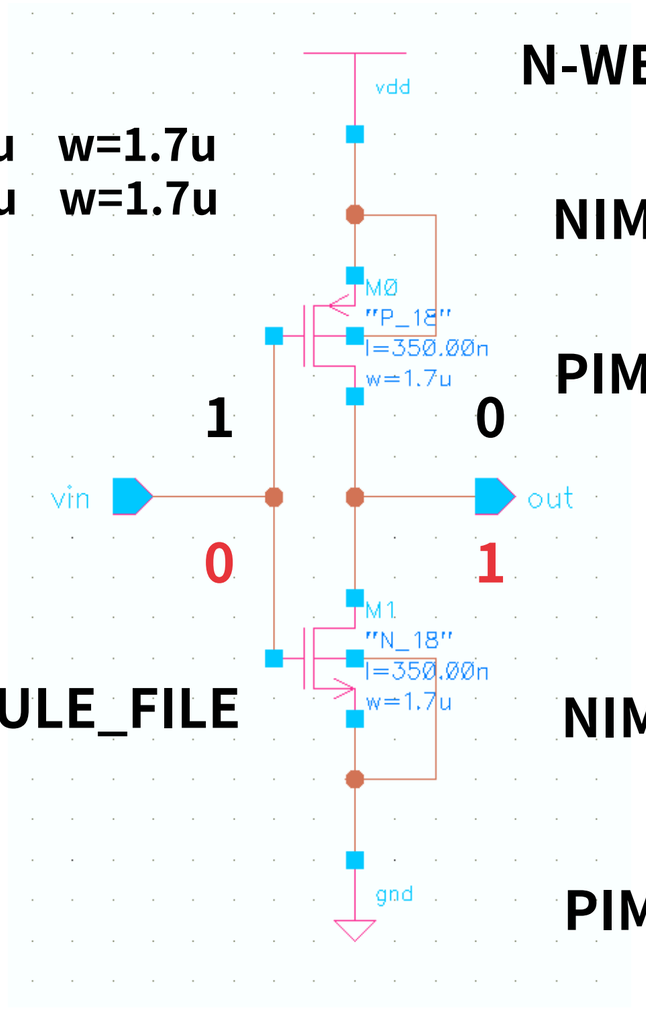
反向器 (Inverter)

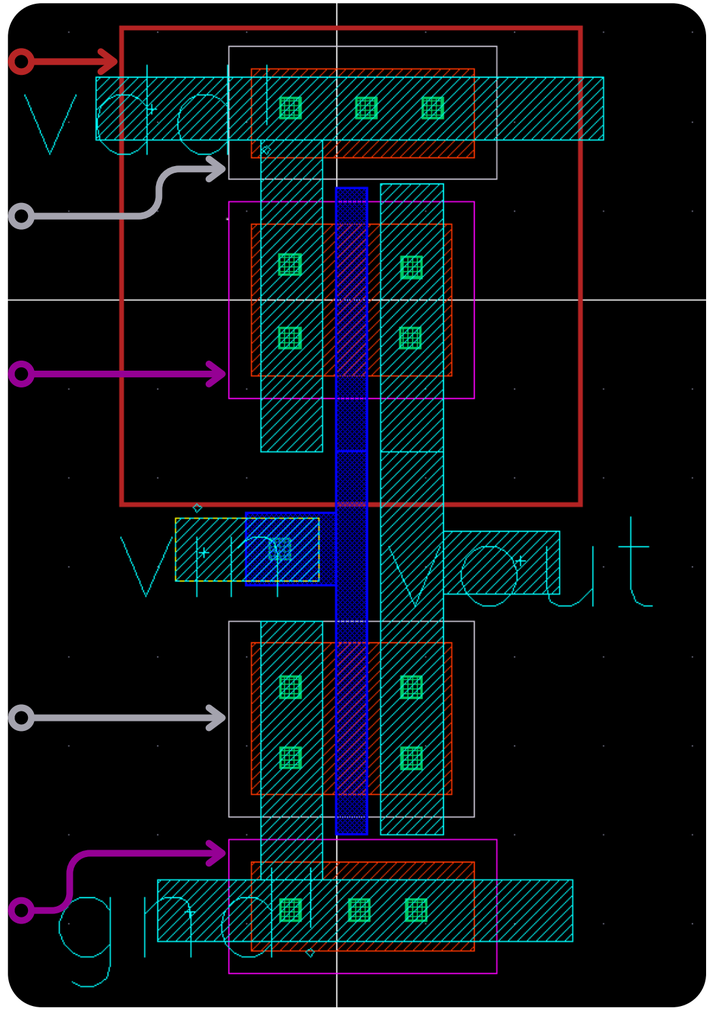
```
.subckt  inv_exp  vin  vout  vdd  gnd
M0  vout  vin  vdd  vdd  p_18  l=0.35u  w=1.7u
M1  vout  vin  gnd  gnd  n_18  l=0.35u  w=1.7u
.ends
```

DRC → LVS → LPE

Qcalibre –lpe [-r|-c|-rc|-rcc] LPE_RULE_FILE

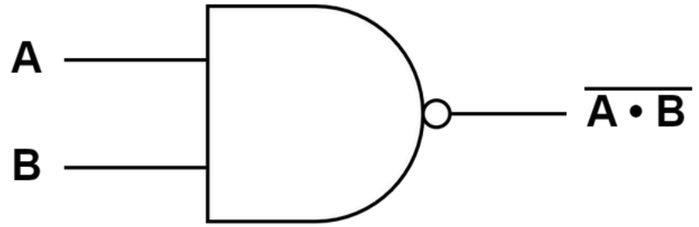
POST SIM: ADS、HSPICE





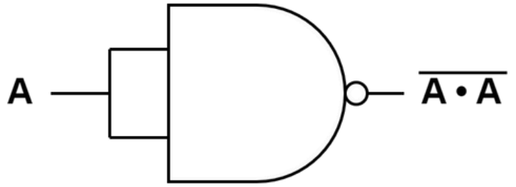
- Computer-Aided Design (電腦輔助電路設計)

萬能的 NAND 閘



A	B	A • B	$\overline{A \cdot B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

NOT



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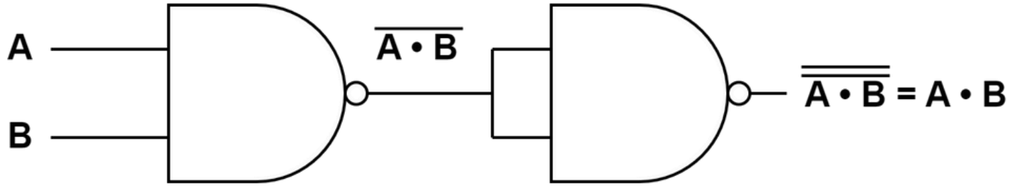
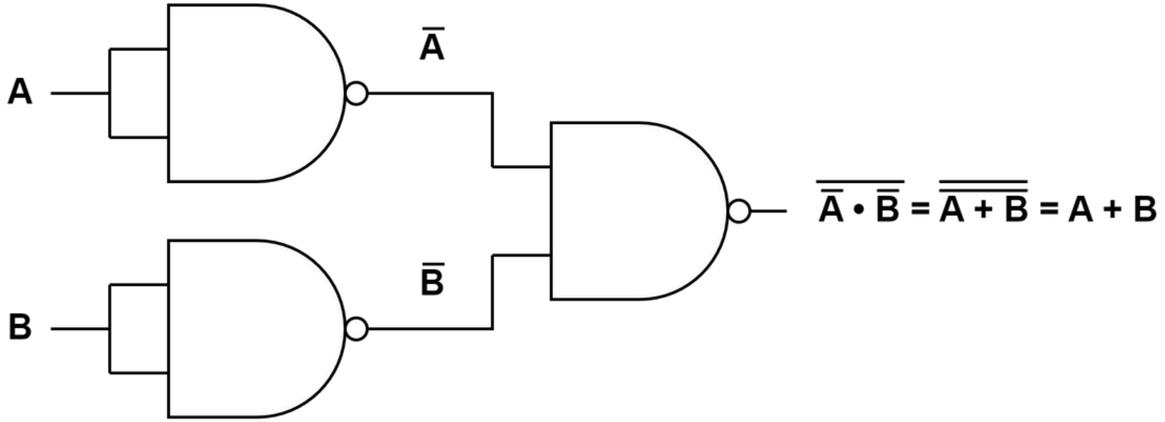


OR



03

類比電路
Analog Circuit

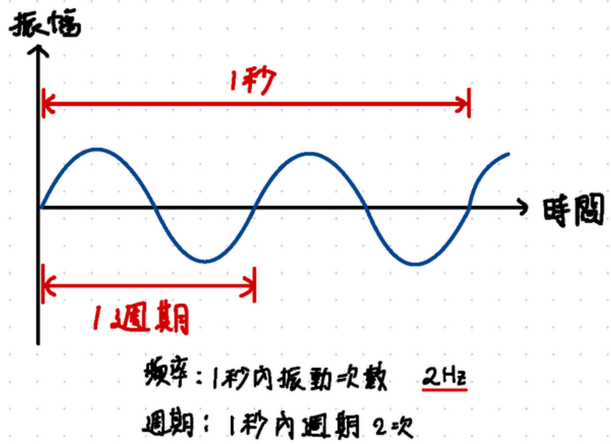
關於類比電路，我知道的事

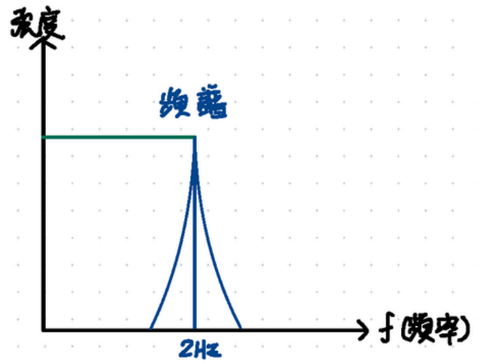
類比電路

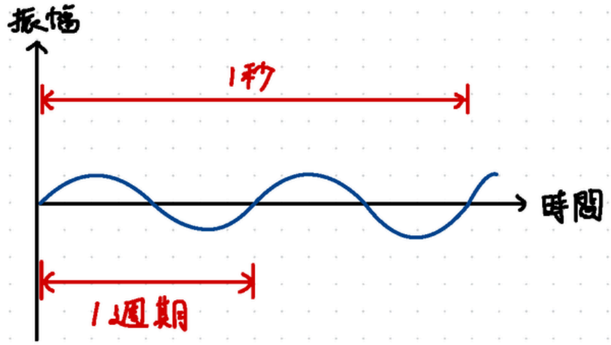
Analog Circuit

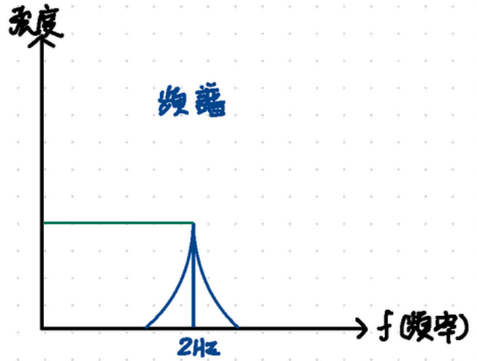
- CMOS Modeling
- Single-ended Amplifier
- Current Mirror & Native Loading
- Differential Amplifier
- Frequency Response
- Feedback System
- Noise
- OP Amplifier
- Stability & Frequency Compensation
- Reference Circuits
- Mismatch & Layout
- Switch Cap & Variable Cap & Oscillator

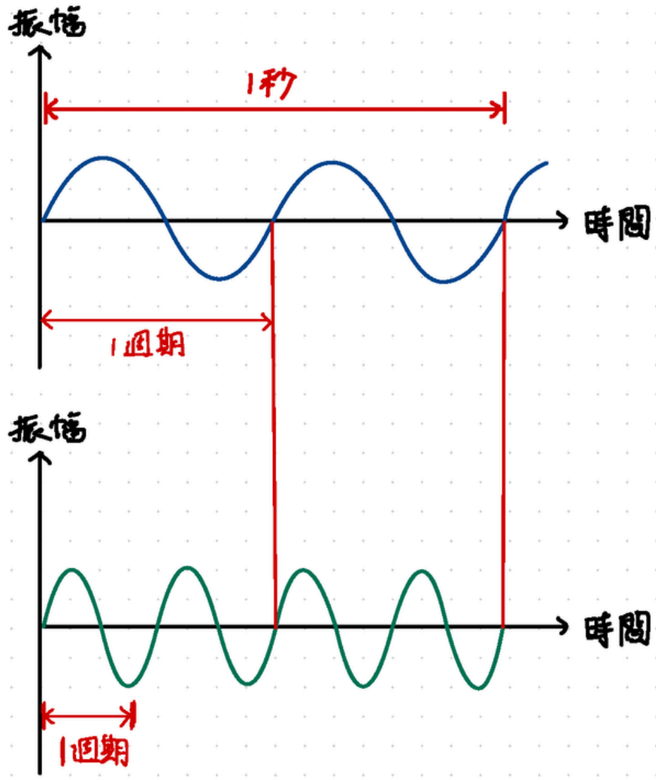
訊號轉換 Signal Translation

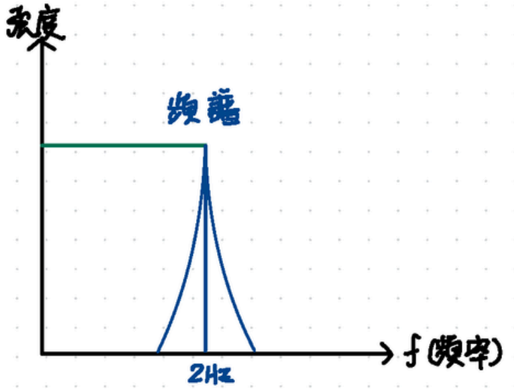


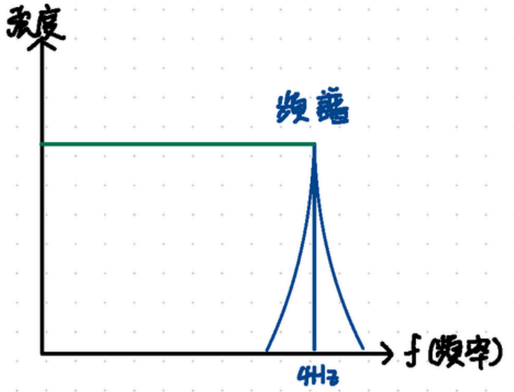




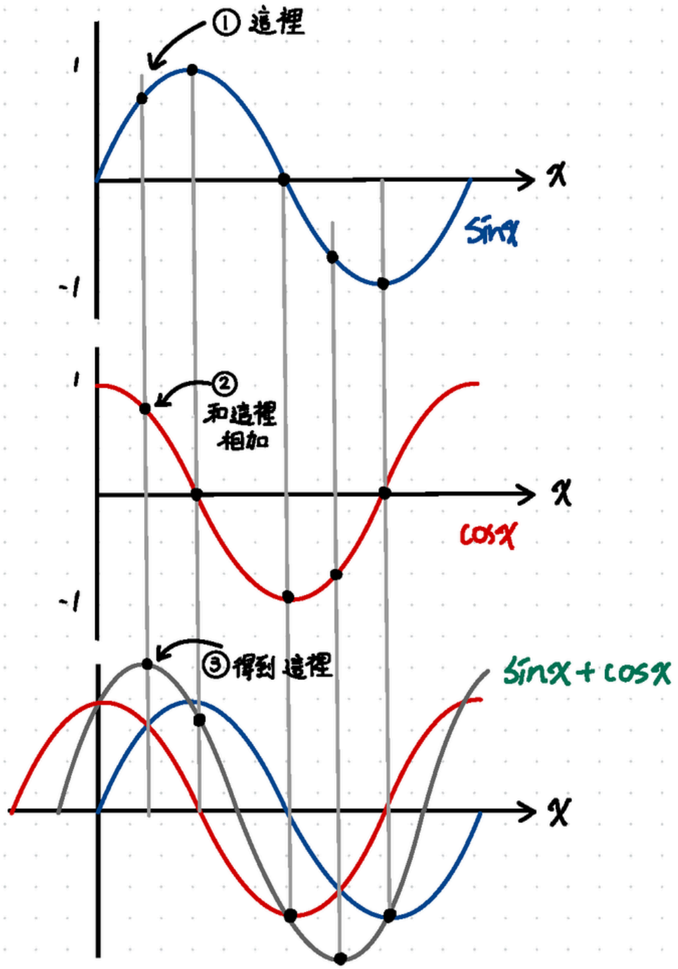


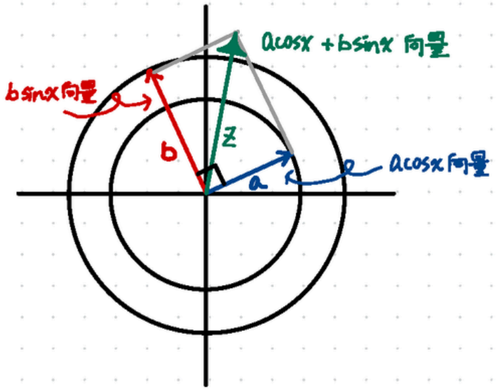


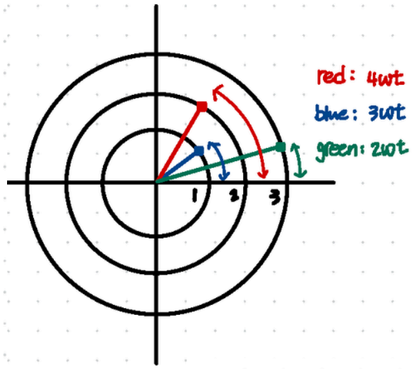


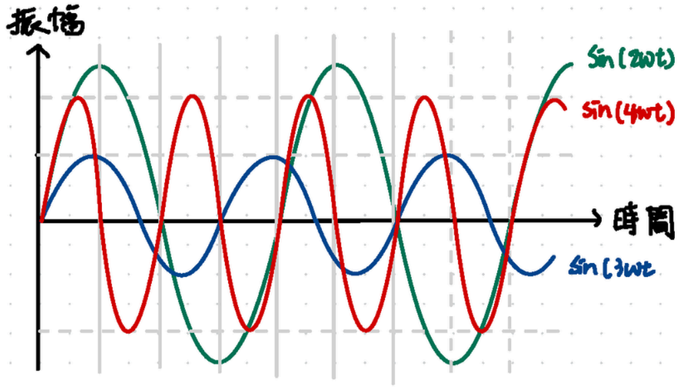


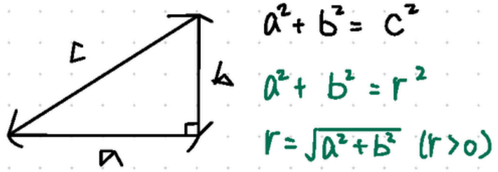
訊號轉換 Signal Translation







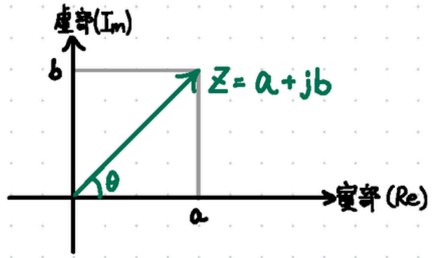


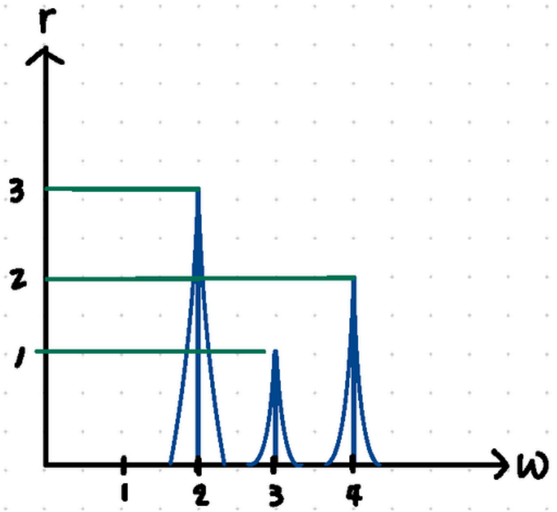


$$a^2 + b^2 = c^2$$

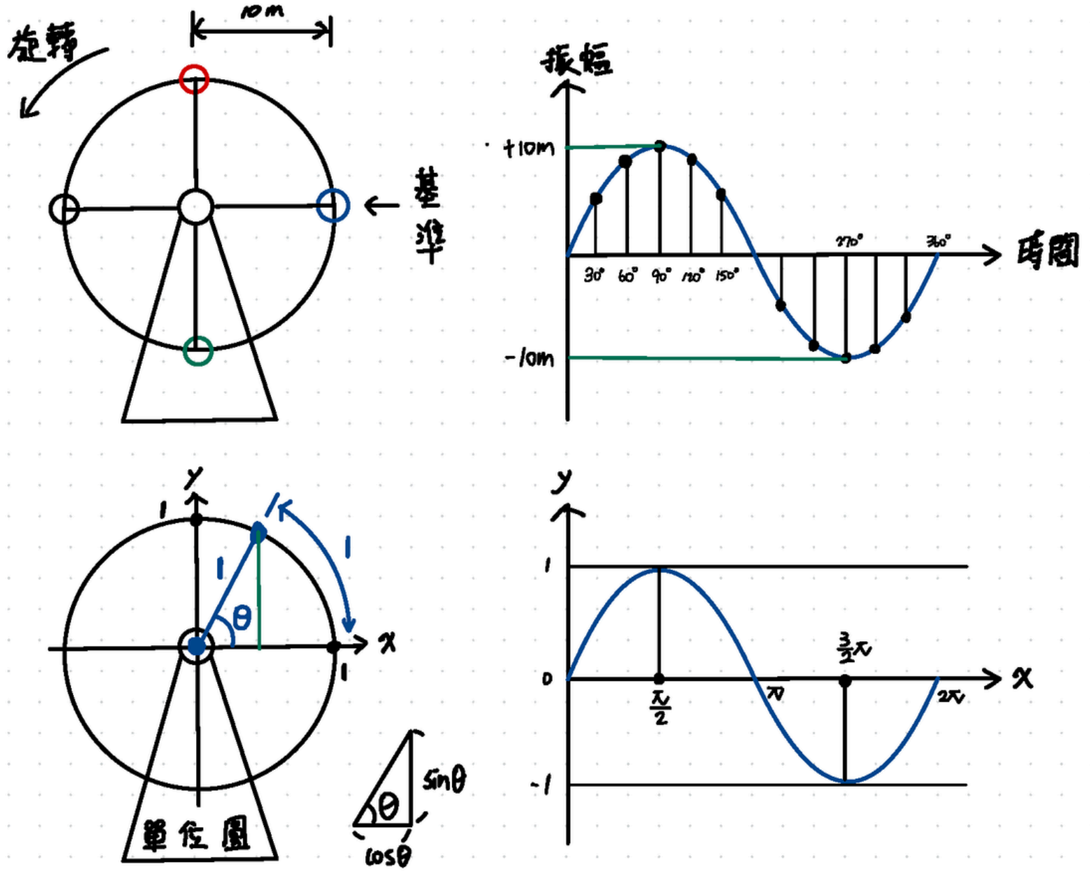
$$a^2 + b^2 = r^2$$

$$r = \sqrt{a^2+b^2} \quad (r>0)$$

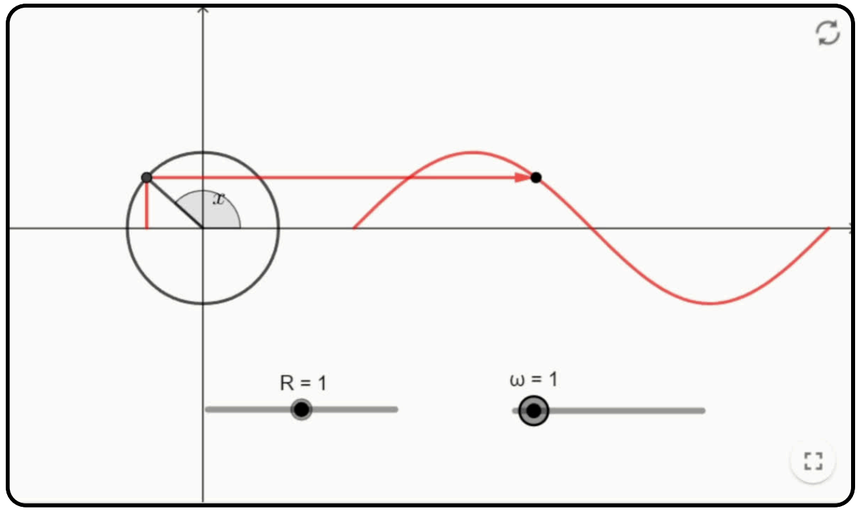


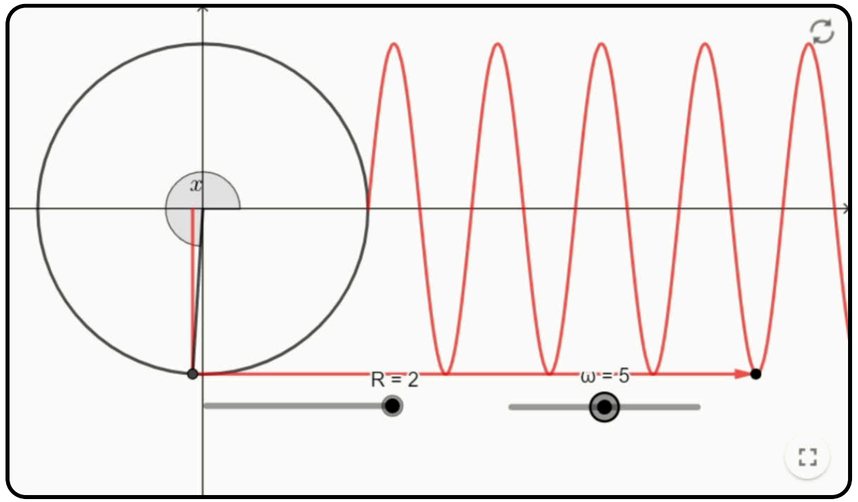


訊號轉換 Signal Translation



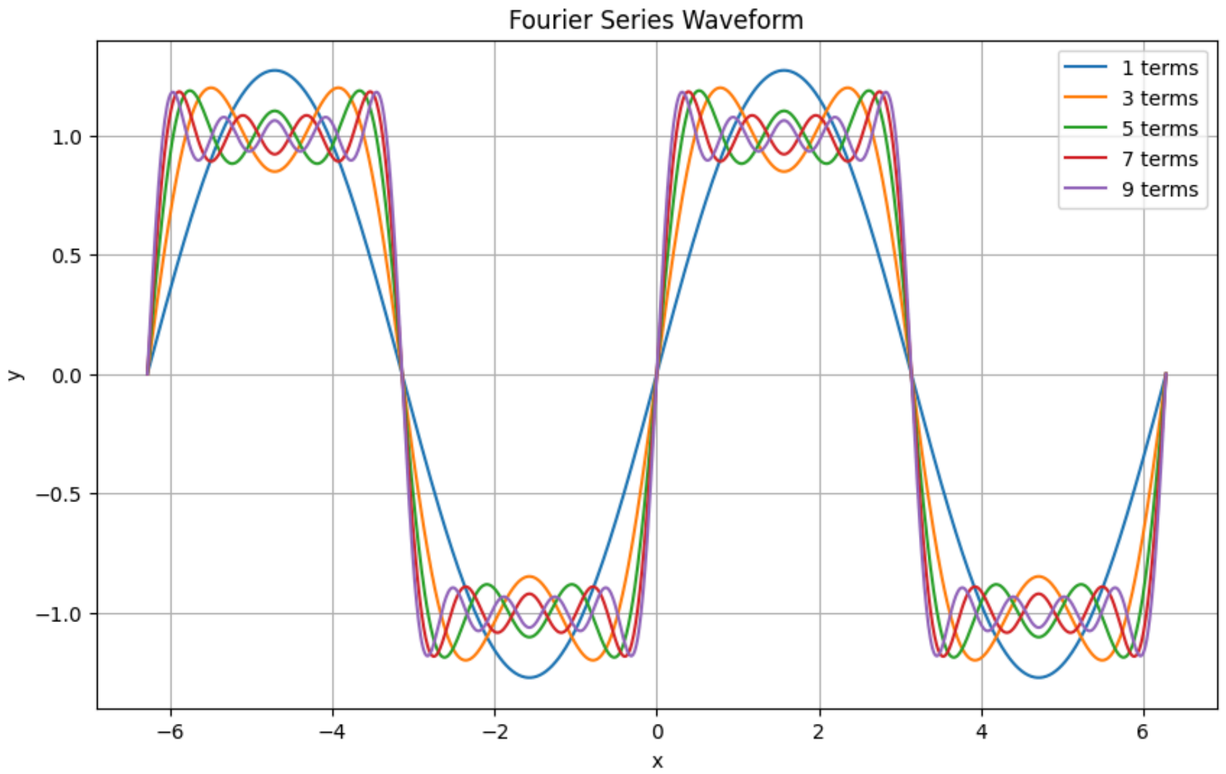
1弧度 (1 rad)：圓周上取一段與半徑等長的弧，所形成的角度。





傅立葉級數 Fourier Series

```python
# Chatgpt 生成
import numpy as np
import matplotlib.pyplot as plt

# 定義傅立葉級數
def fourier_series(x, n_terms):
    result = 0
    for n in range(1, n_terms + 1):
        # Use odd terms
        if n % 2 != 0:
            result += (4 / (np.pi * n)) * np
.sin(n * x)
    return result

# 生成波型
x_values = np.linspace(-2 * np.pi, 2 * np.pi,
1000)

plt.figure(figsize=(10, 6))
for n_terms in range(1, 10, 2):
    y_values = fourier_series(x_values, n_terms
)
    plt.plot(x_values, y_values, label=f'{
n_terms} terms')

plt.title('Fourier Series Waveform')
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True)
plt.show()
```

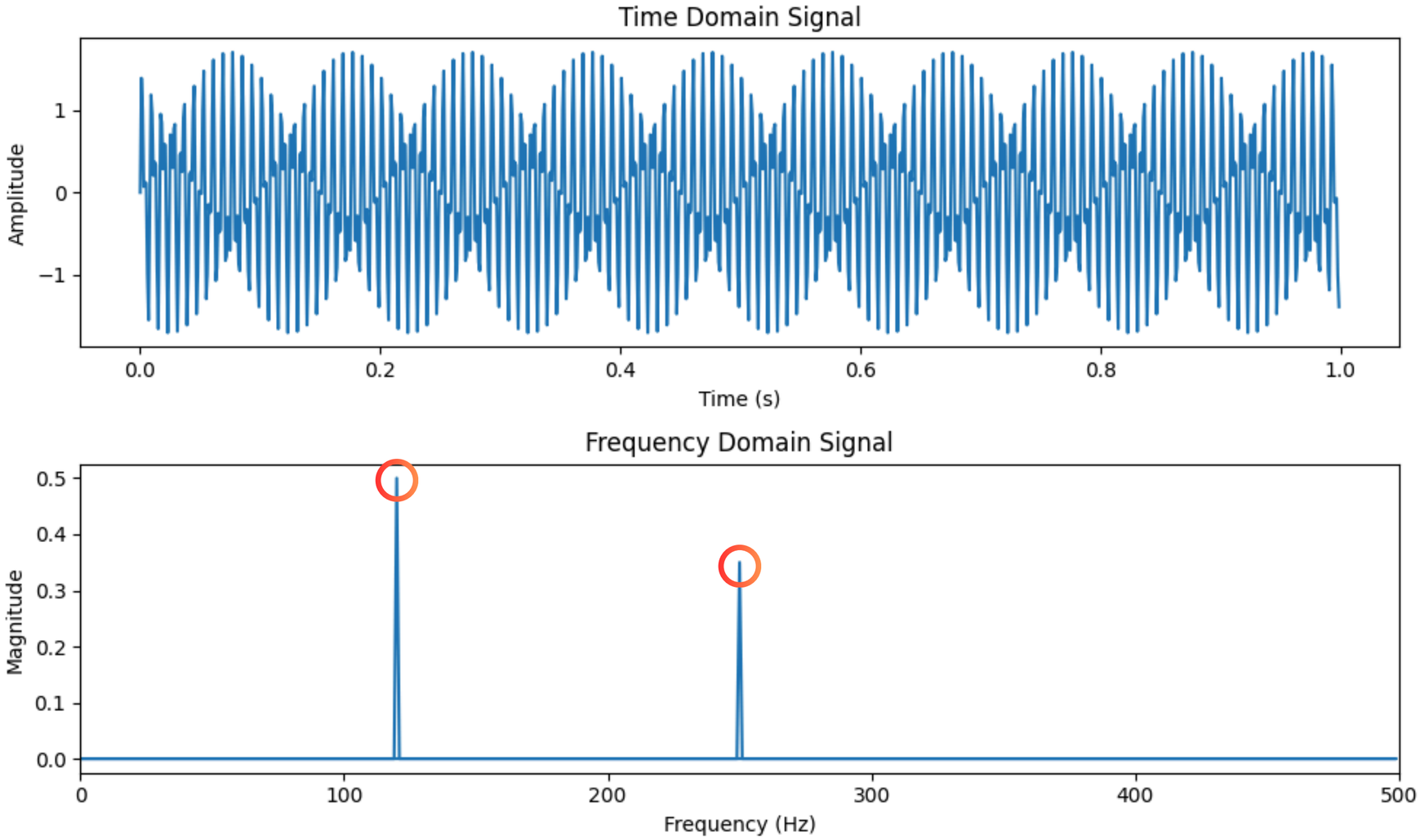


$$F(x) = \frac{1}{2}a_0 + a_1\sin(x) + a_2\sin(2x) + a_3\sin(3x) + \cdots + a_n\sin(nx) + \cdots$$

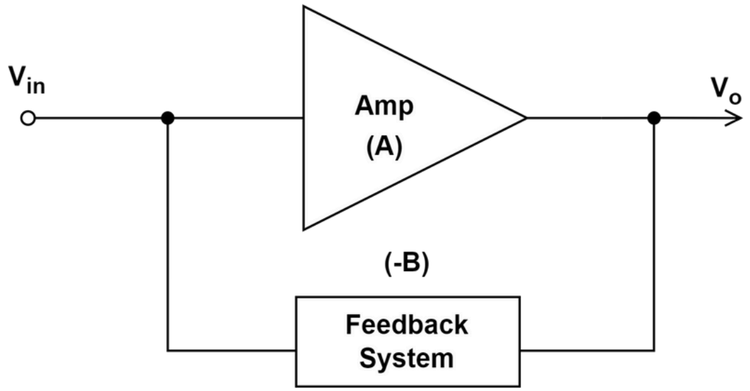
$$= \frac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n \sin(nx))$$

傅立葉轉換 Fourier Transform

```python
# Chatgpt 生成
import numpy as np
import matplotlib.pyplot as plt

# 產生一個例子時域訊號，例如一個具有兩個頻率分量的信號
# 這裡我們產生一個由兩個正弦波組成的信號
Fs = 1000  # 取樣頻率 (Hz)
t = np.arange(0, 1, 1/Fs)
# 時間向量，從0到1秒，以1/Fs的步長取樣
f1 = 250  # 第一個正弦波的頻率 (Hz)
f2 = 120  # 第二個正弦波的頻率 (Hz)
x = 0.7*np.sin(2*np.pi*f1*t) + np.sin(2*np.pi*f2*t)
# 合成信號

# 繪製時域訊號
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.plot(t, x)
plt.title('Time Domain Signal')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')

# 執行傅立葉變換
X = np.fft.fft(x)  # 執行傅立葉變換

# 計算對應的頻率向量
frequencies = np.fft.fftfreq(len(x), 1/Fs)

# 計算振幅並進行歸一化
magnitude = np.abs(X) / len(x)

# 繪製頻域訊號
plt.subplot(2, 1, 2)
plt.plot(frequencies, magnitude)
plt.title('Frequency Domain Signal')
plt.xlabel('Frequency (Hz)')
plt.ylabel('Magnitude')
plt.xlim(0, Fs/2)  # 只繪製正頻率部分
plt.tight_layout()
plt.show()
```



振盪器 Oscillator



▲ Negative feedback amplifier (負回授)

$$A(V_{in} - BV_o) = V_o$$

$$AV_{in} - ABV_o = V_o$$

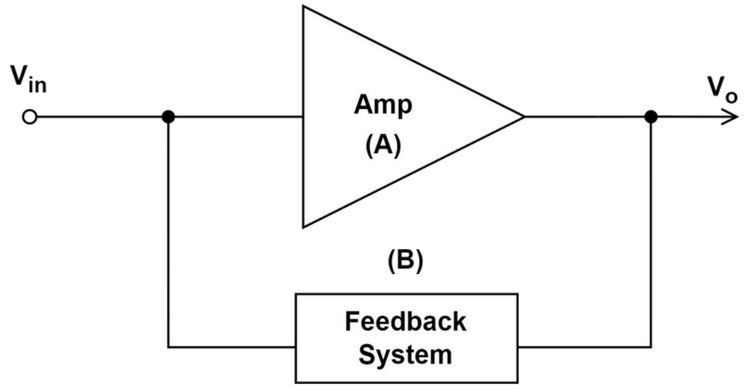
$$AV_{in} = ABV_o + V_o$$

$$AV_{in} = (1 + AB)V_o$$

$$V_o = \frac{AV_{in}}{(1 + AB)}$$

$$A(\text{放大率}) = \frac{V_o}{V_{in}} = \frac{A}{1 + AB}$$

- 減少雜訊
- 放大率降低
- 擴大頻寬(領域)
- 穩定放大率(T、Vdd)影響
- 改變輸出入阻抗



▲ Positive feedback amplifier (正回授)

$$A(\text{放大率}) = \frac{V_o}{V_{in}} = \frac{A}{1 - AB}$$

$AB = 1$時，$A(\text{放大率}) = \infty$(無限大)

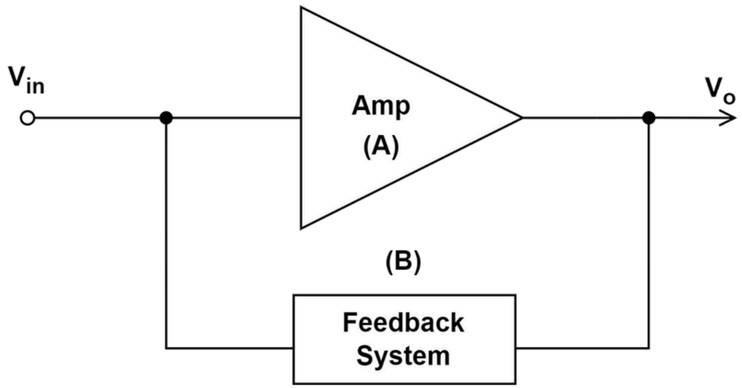
稱為迴路增益(loop gain)

- 必須具有正回授
- 放大率要足夠
- 符合巴克豪生準則
- 範圍實際最大Vdd

NTHU 實驗教學物理實驗室: http://www.phys.nthu.edu.tw/~exphy/

. DeltaMOOCx-基本振盪電路: https://www.youtube.com/watch?v=eYWXjYhNTtw&ab_channel=.DeltaMOOCx/

振盪器 Oscillator

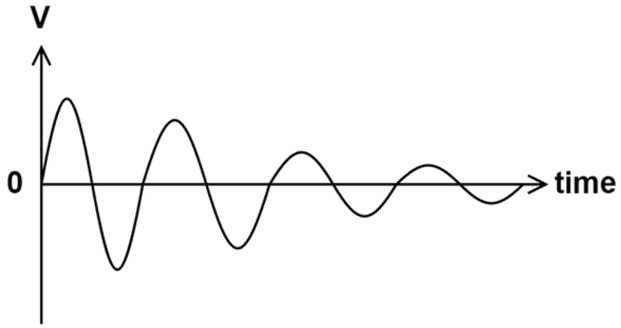


▲ positive feedback amplifier (正回授)

$$A(\text{放大率}) = \frac{V_o}{V_{in}} = \frac{A}{1 - AB}$$

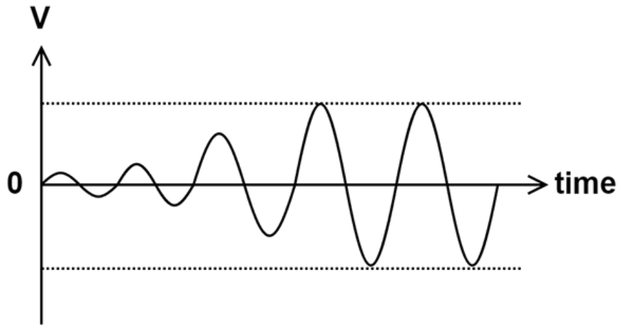
$AB = 1$時，$A(\text{放大率}) = \infty(\text{無限大})$

稱為迴路增益(loop gain)



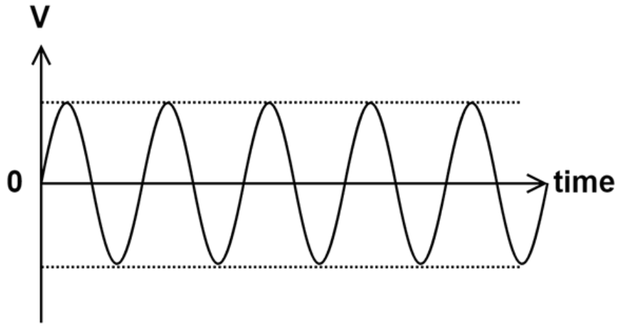
$BA < 1\,, \angle 0^\circ$

減幅震盪



$BA > 1\,, \angle 0^\circ$

增幅震盪



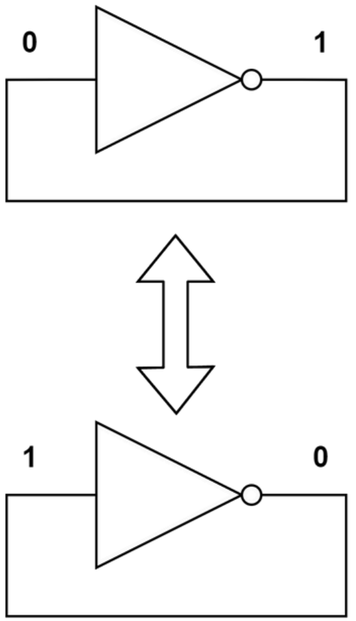
$BA = 1\,, \angle 0^\circ$

等幅震盪

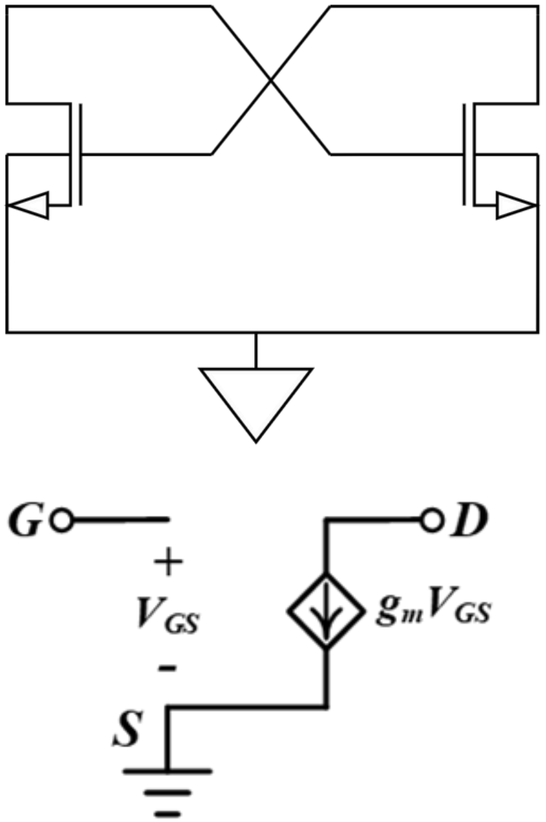
NTHU 實驗教學物理實驗室: http://www.phys.nthu.edu.tw/~exphy/

. DeltaMOOCx-基本振盪電路: https://www.youtube.com/watch?v=eYWXjYhNTtw&ab_channel=.DeltaMOOCx/

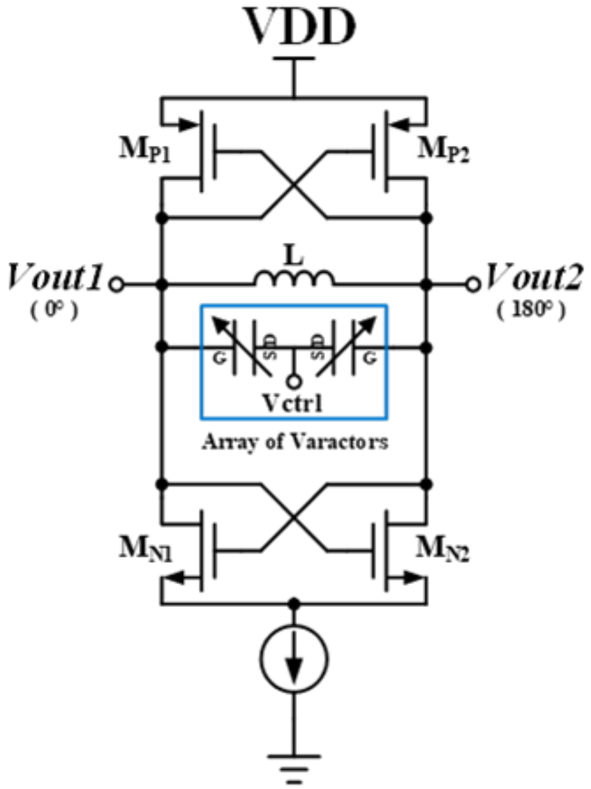
振盪器 Oscillator



▲ 概念圖

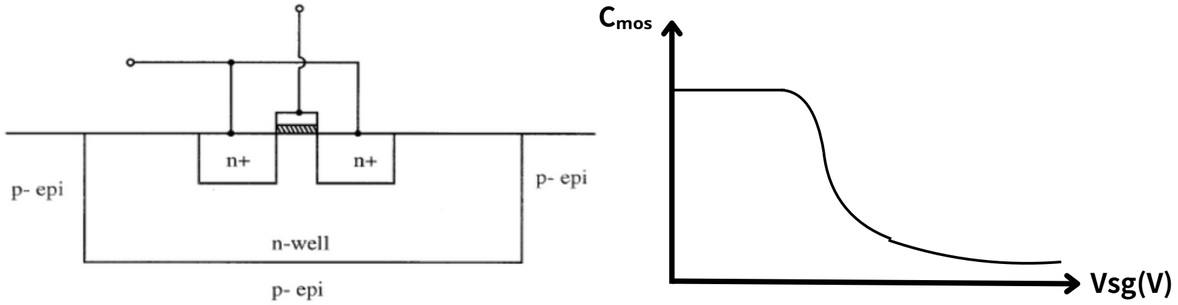


▲ Cross coupled oscillator

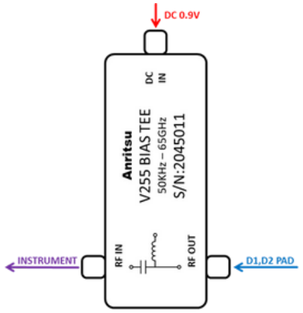


▲ LC-Tank oscillator

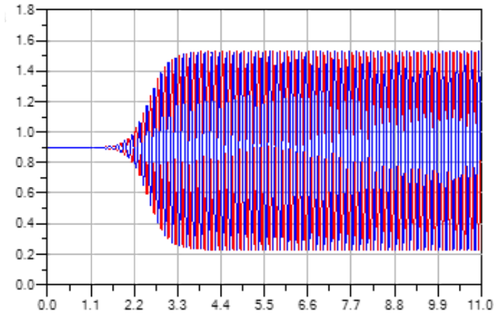
壓控振盪器 VCO

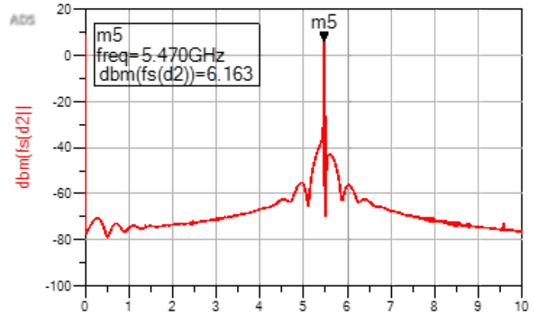


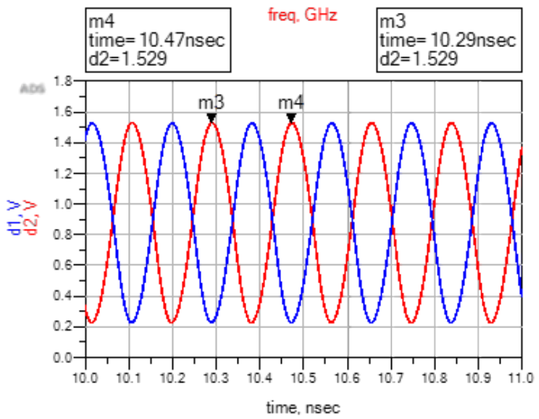
▲ Accumulation-mode MOS varactor

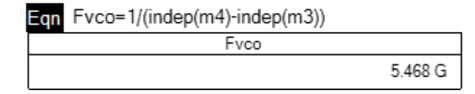


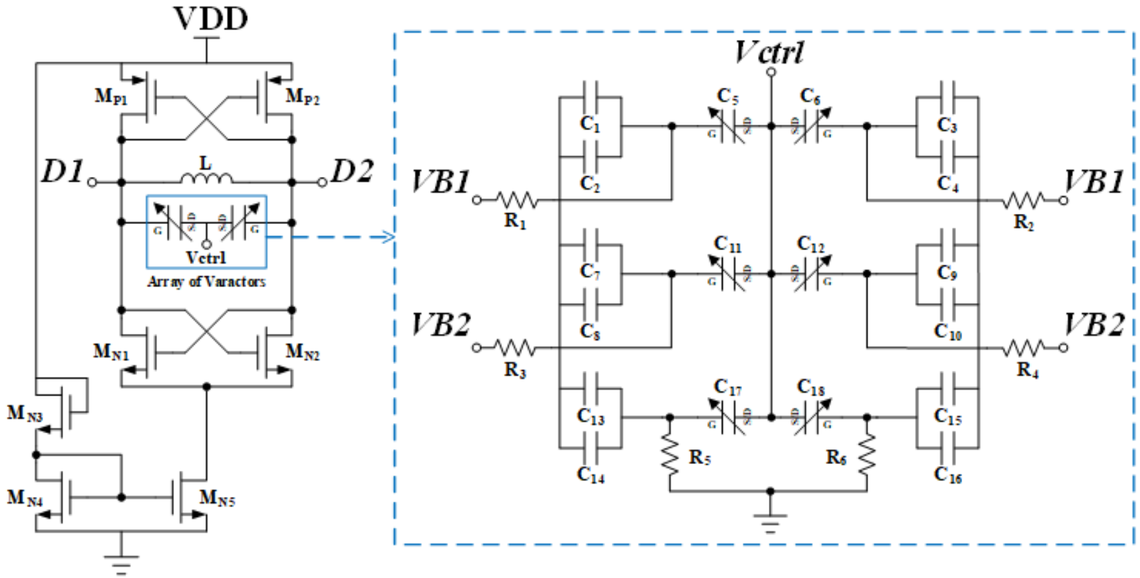
▲ Bias Tee



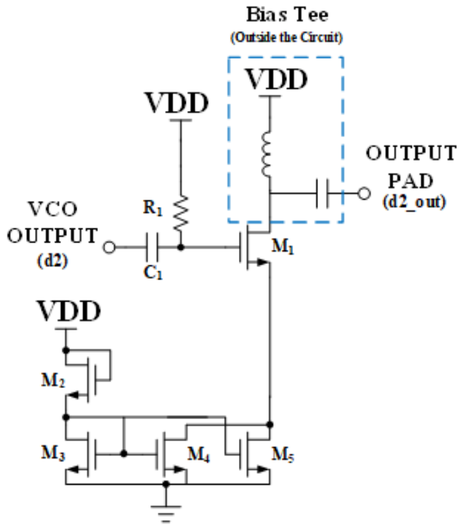








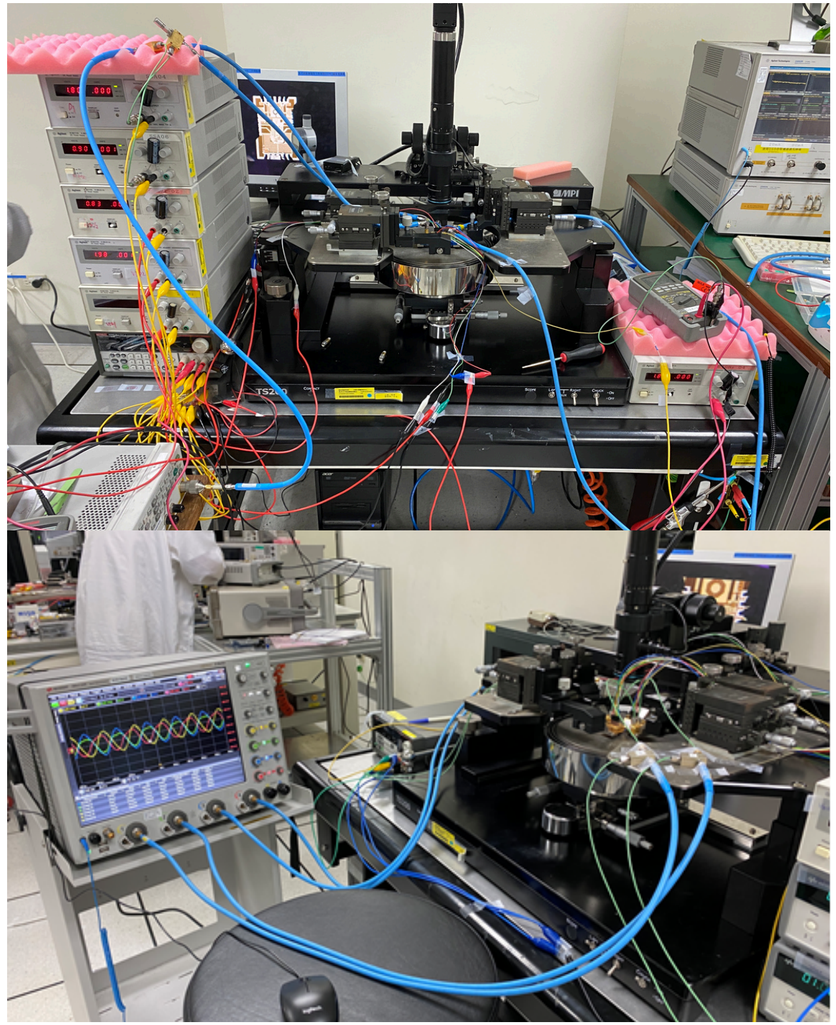
▲ Structure of voltage-controlled oscillator

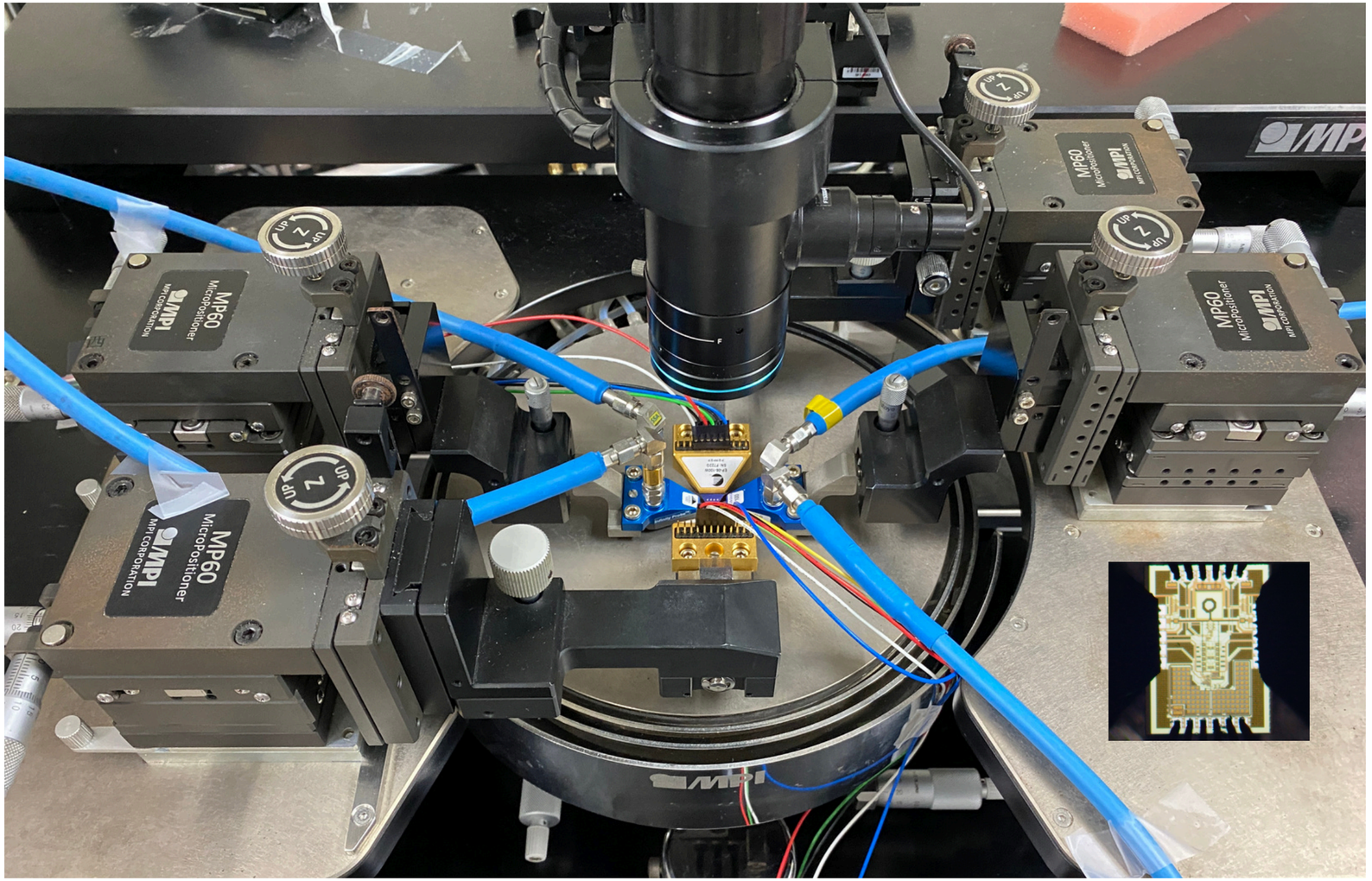


▲ Buffer

MOS Varactor : 鍾岳翰,基於電壓控制的連續性可變電容之研究,國立雲林科技大學研究所碩士論文,民國103年。

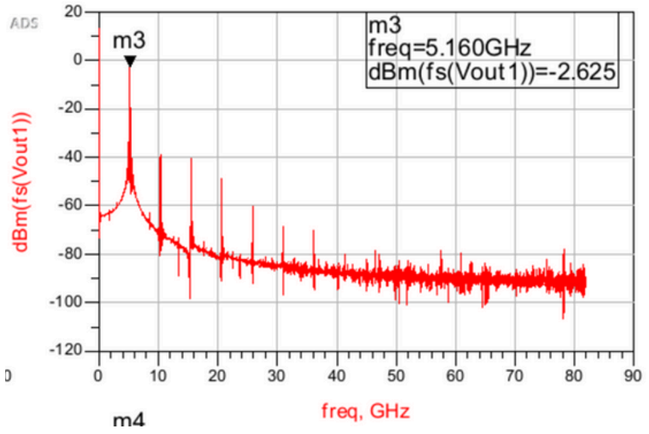
實際量測 Measurement

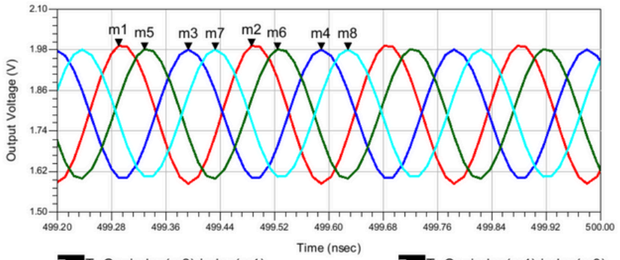




▲ On Wafer Measurement

實際量測 Measurement





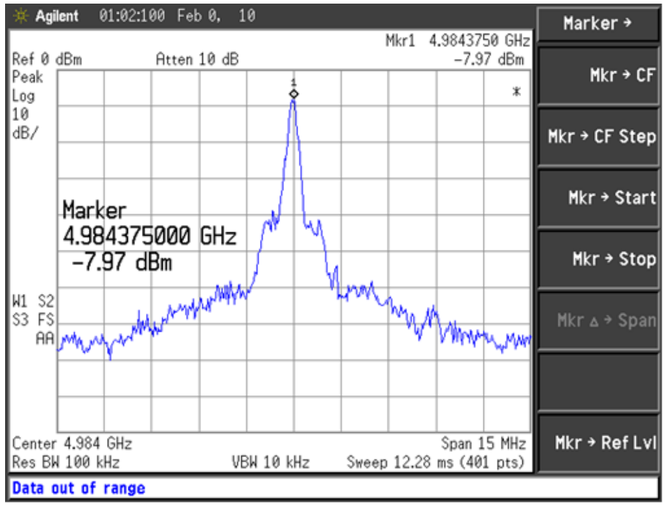
Eqn T_Qp=indep(m2)-indep(m1)
Eqn F_Qp=1/T_Qp
Eqn T_Ip=indep(m6)-indep(m5)
Eqn F_Ip=1/T_Ip
Eqn P1=((indep(m5)-indep(m1))/1.9e-10)*360
Eqn P2=((indep(m3)-indep(m1))/1.9e-10)*360
Eqn P3=((indep(m7)-indep(m1))/1.9e-10)*360

Eqn T_Qn=indep(m4)-indep(m3)
Eqn F_Qn=1/T_Qn
Eqn T_In=indep(m8)-indep(m7)
Eqn F_In=1/T_In

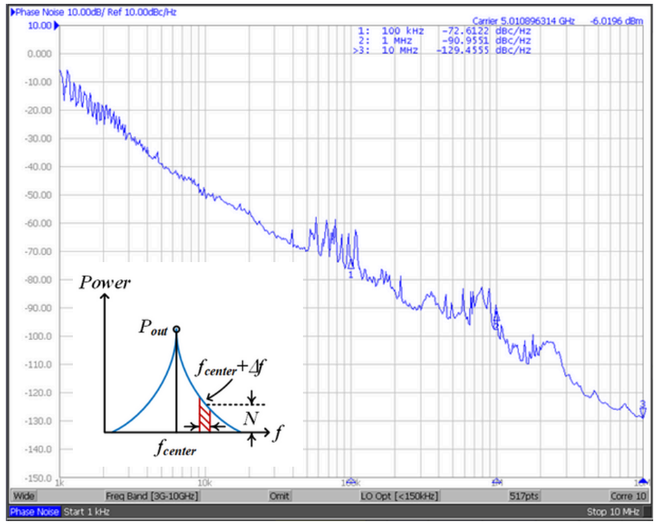
T_Qp	F_Qp	T_Qn	F_Qn	T_Ip	F_Ip	T_In	F_In
195.7 p	5.109 G	195.7 p	5.109 G	195.7 p	5.109 G	195.7 p	5.109 G

P1	P2	P3
72.377	194.425	269.247

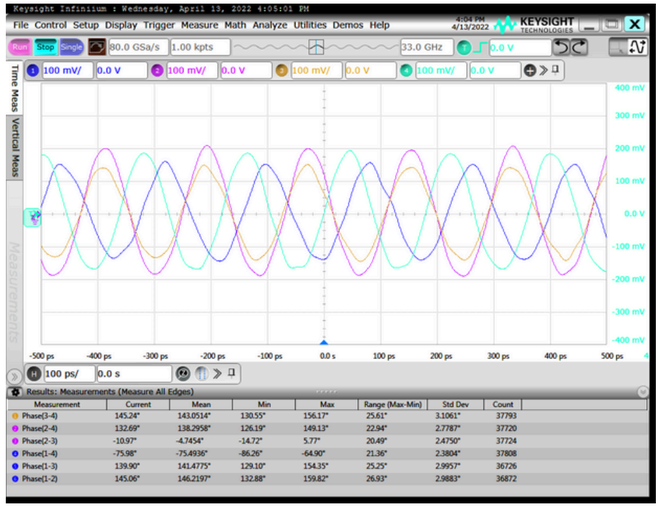
▲ Post-Simulation



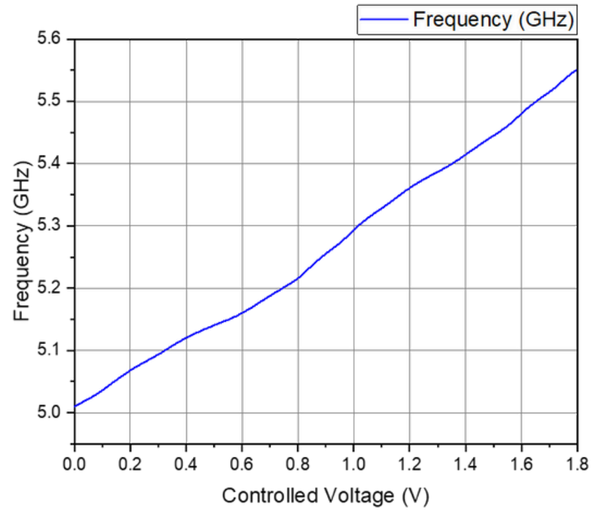
▲ 頻譜分析儀訊號輸出圖



▲ 相位雜訊



▲ 訊號輸出圖



▲ 可操作頻率測試

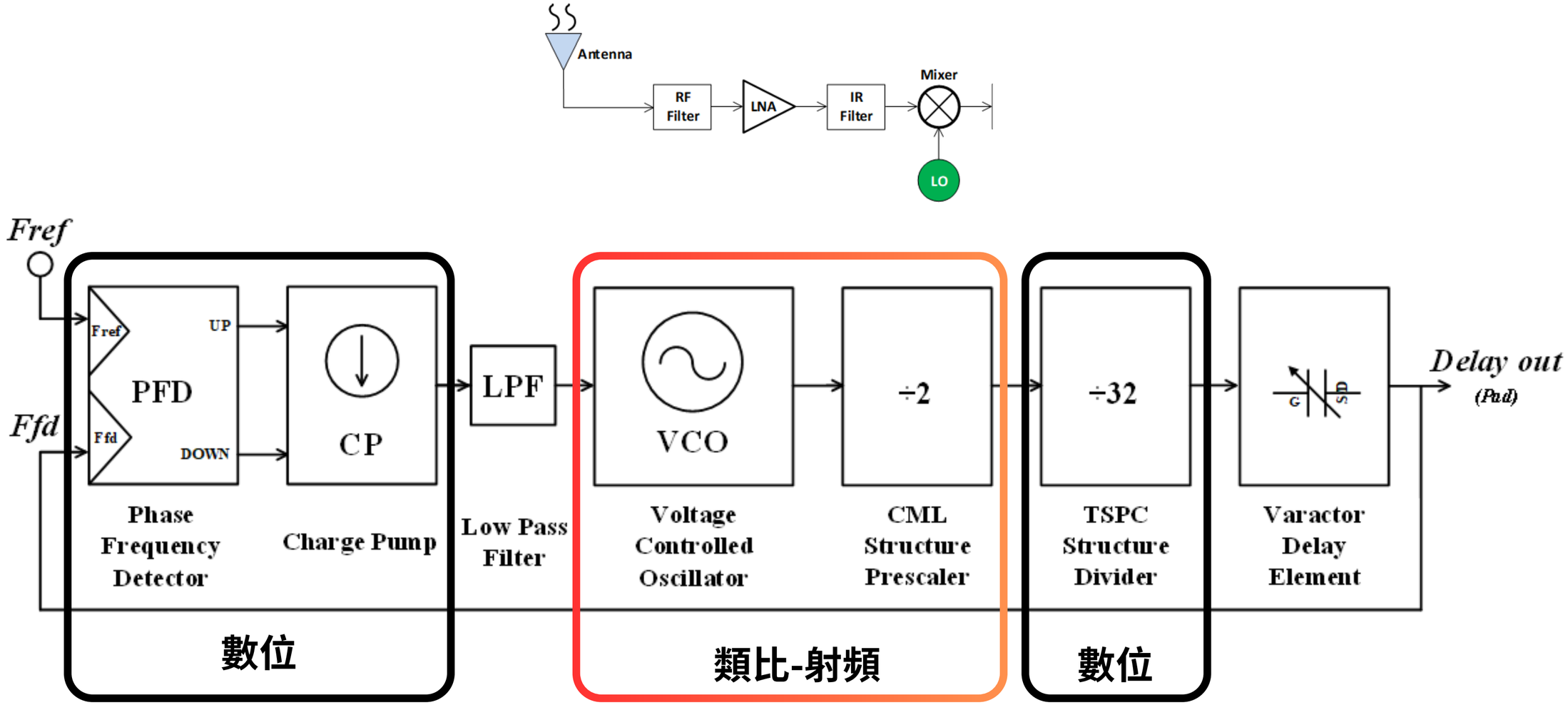
04

總結

Summary

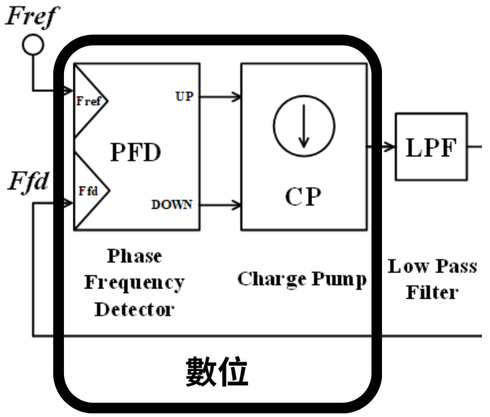
聽到這邊真是辛苦大家~

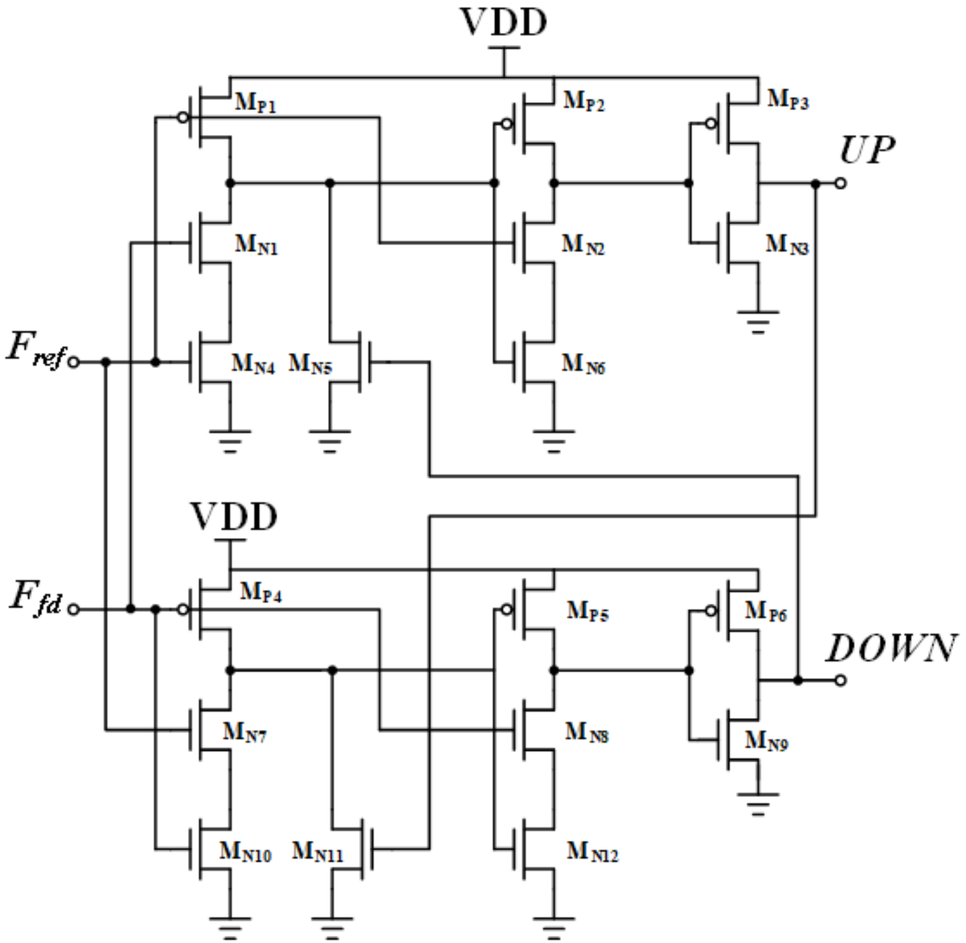
鎖相迴路 Phase-Locked Loop



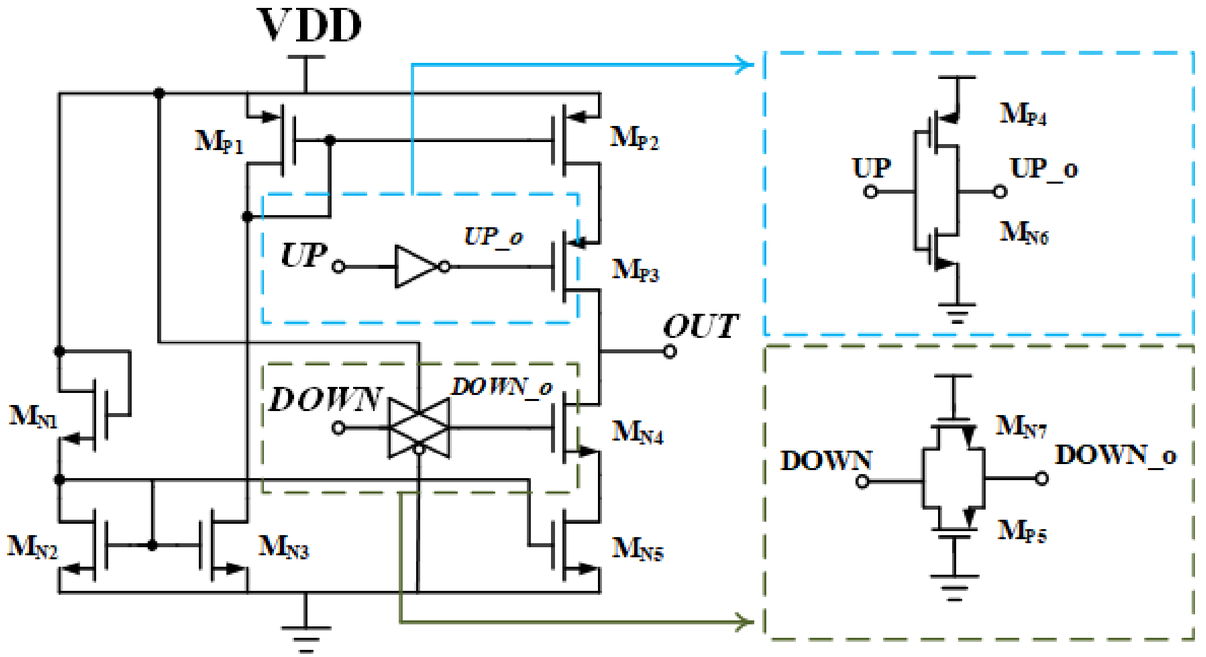
▲ 鎖相迴路系統架構圖

PFD + CP 結構



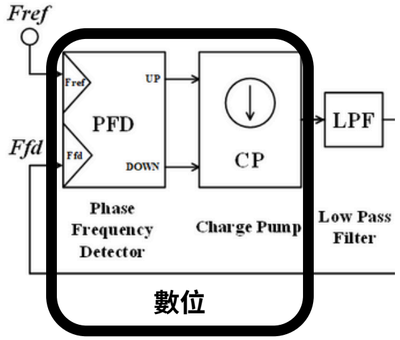


▲ 相位偵測器 (PFD)

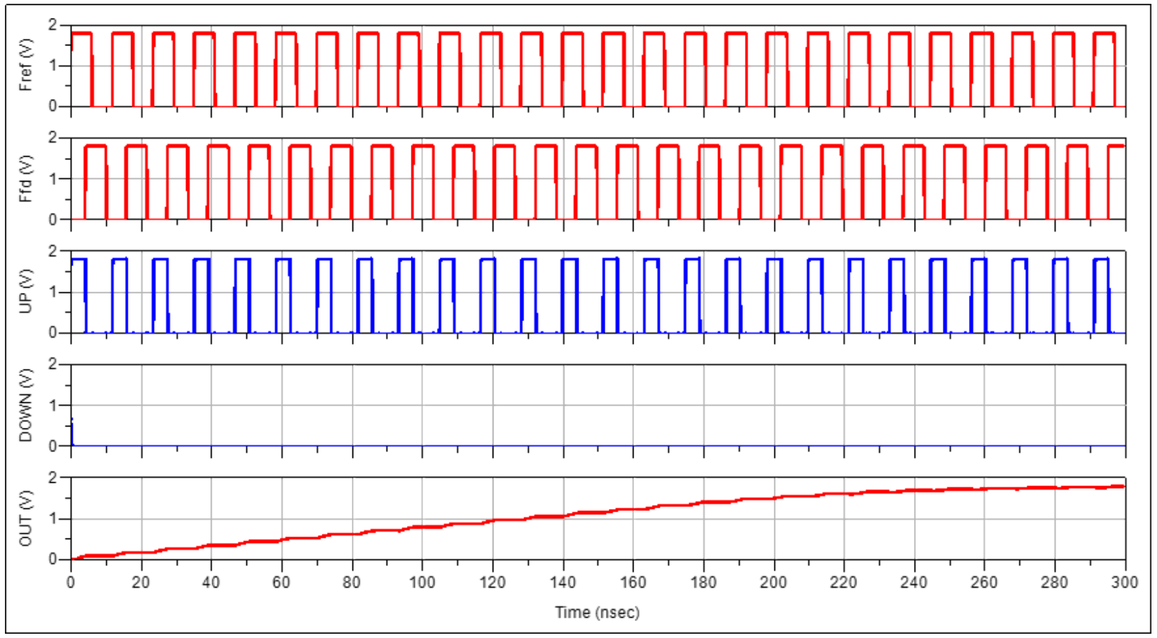


▲ 充電幫補 (CP)

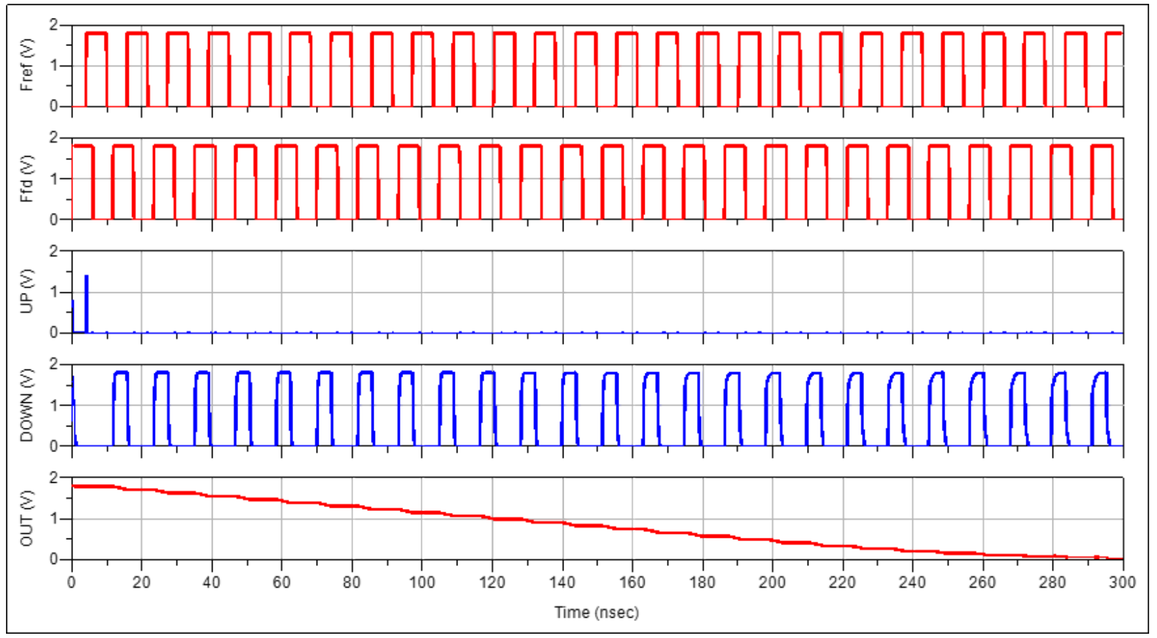
PFD + CP 輸出



相位領先，CP充電時

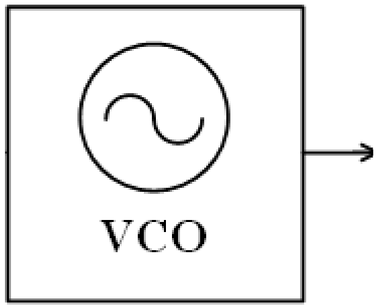


相位落後，CP放電時

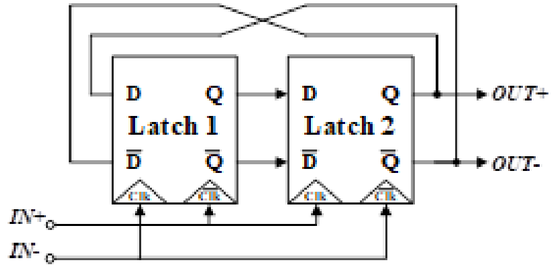


▲ 相位偵測器 (PFD) + 充電幫補 (CP) 輸出結果

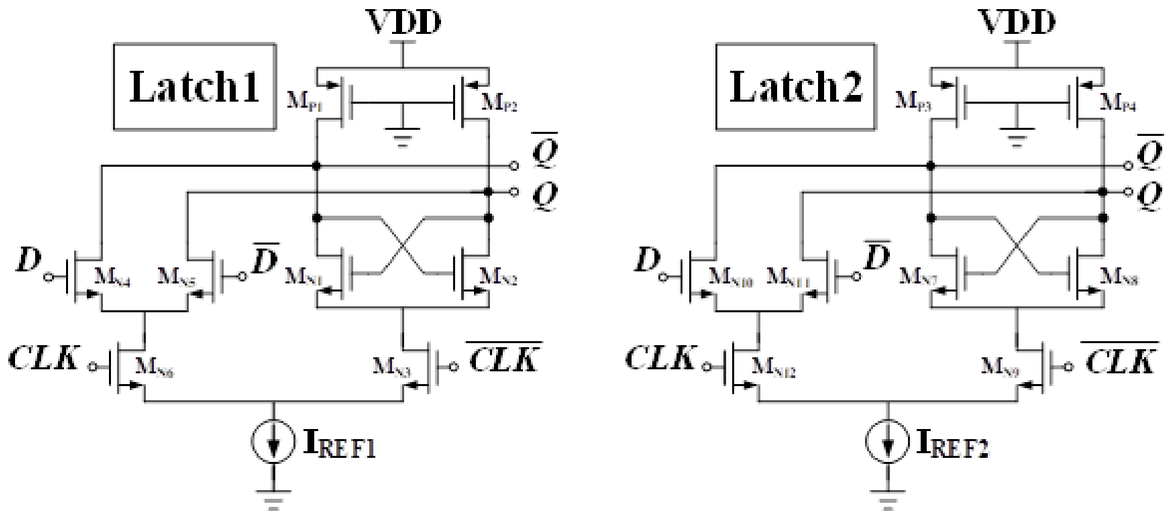
VCO + Prescaler



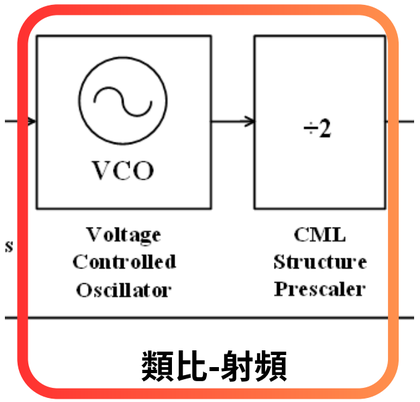
Voltage Controlled Oscillator

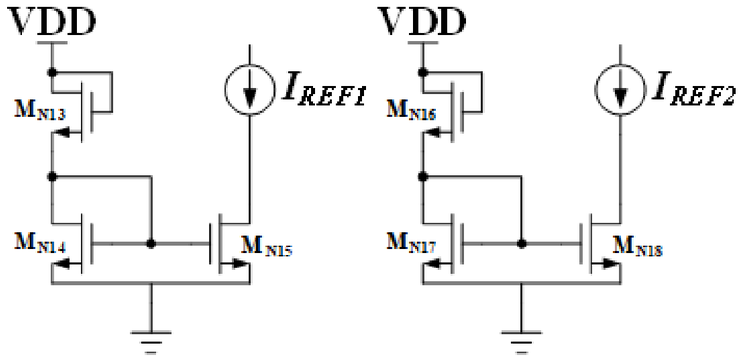


▲ 前置除頻器 Prescaler 簡圖



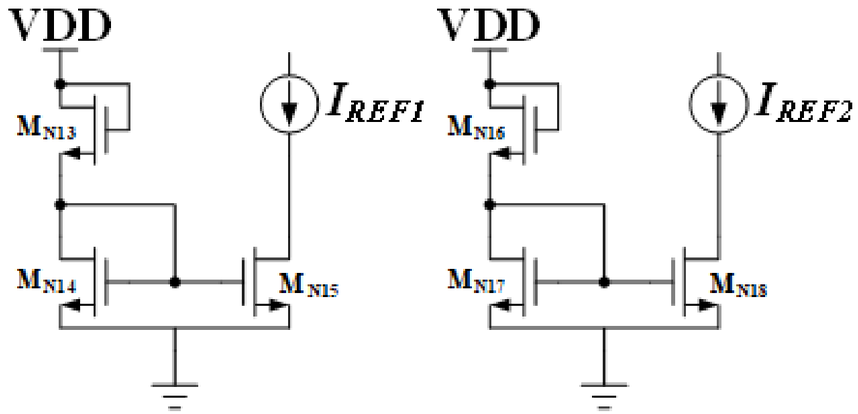
▲ 前置除頻器 Prescaler 內部結構



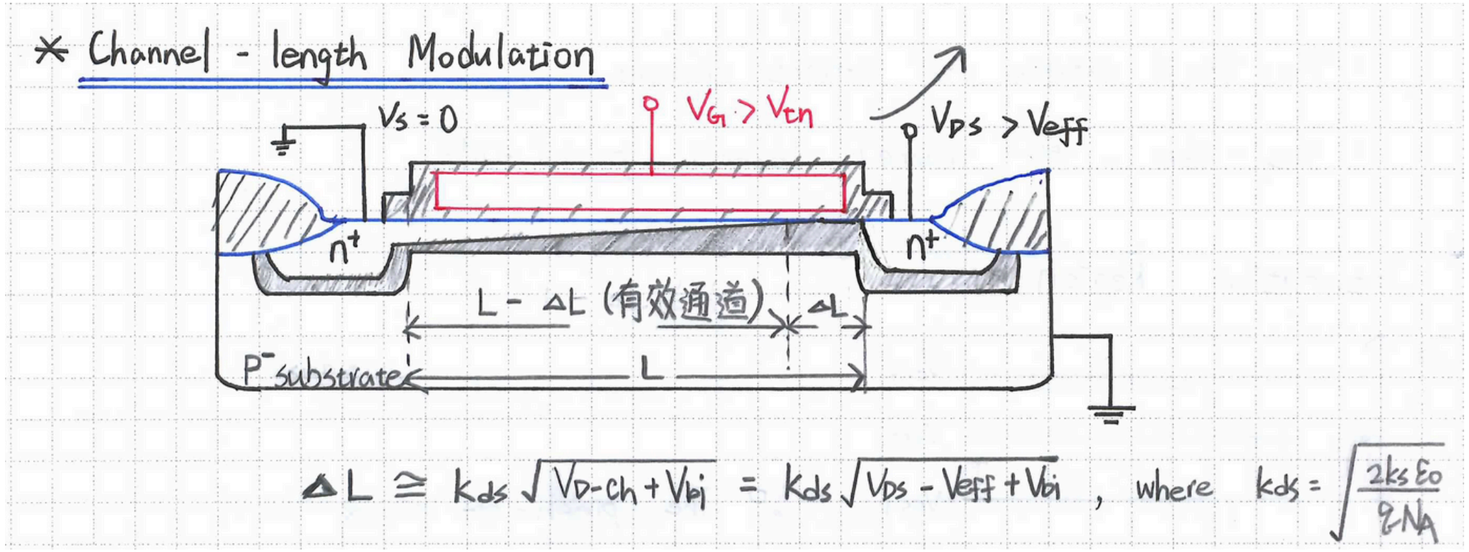


▲ 電流鏡

Channel Length Modulation Effect



▲ 電流鏡



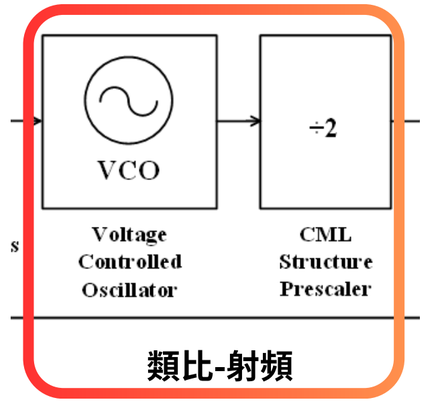
▲ 通道長度調變效應

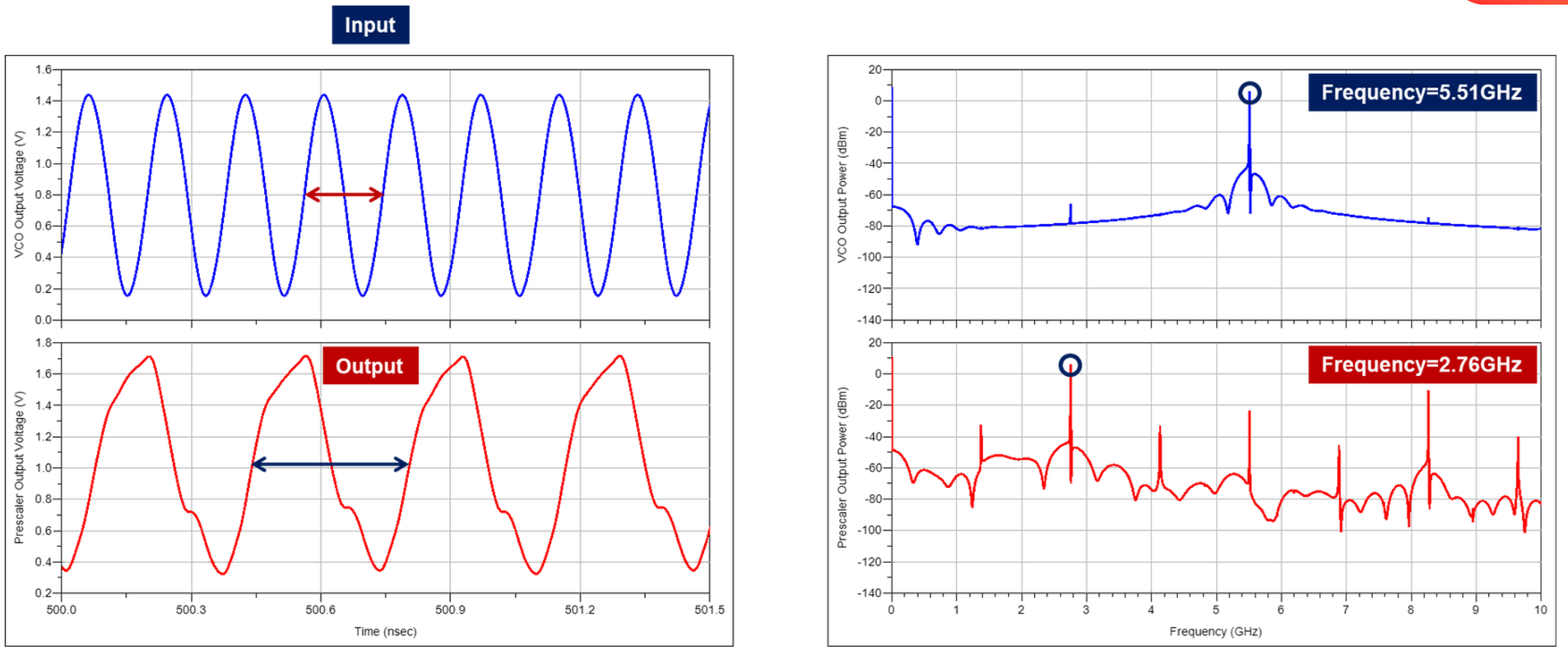
Vds增加，縮小D端空乏區大小，載子通過時間下降，降低有效的通道長度，短通道長度會造成較高的電流。

$$I_D \approx \frac{1}{2}\mu_n Cox\left(\frac{W}{L}\right)(V_{GS}-V_{th})^2(1+\lambda V_{DS})$$

→ 避免使用過小 Length (Gate大小)

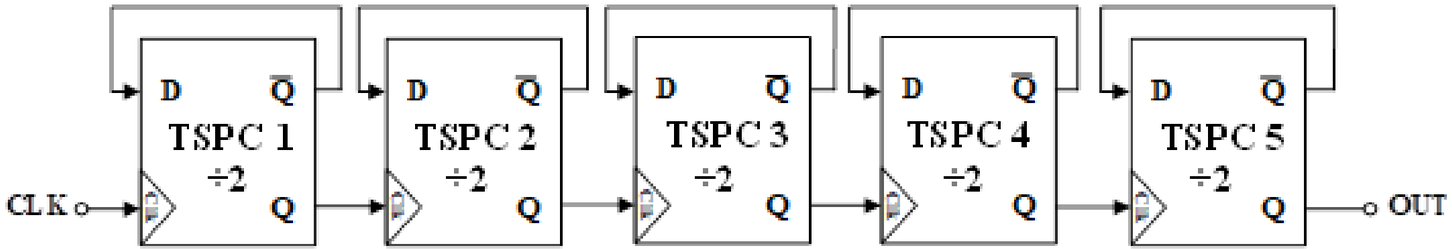
VCO + Prescaler 輸出

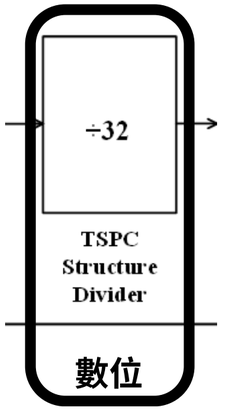


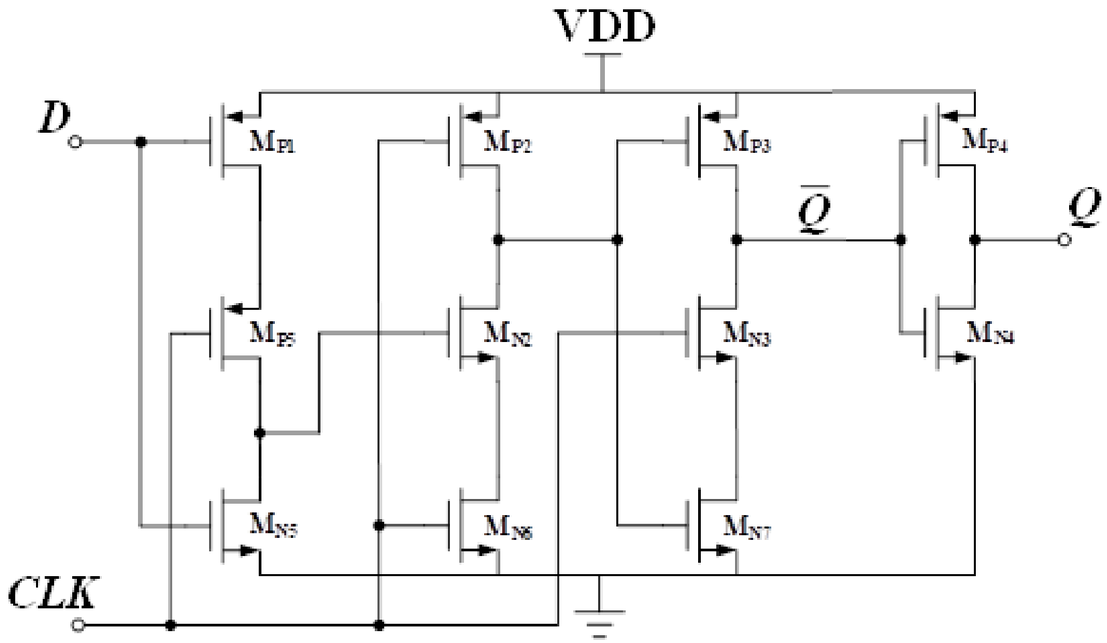


▲ 壓控振盪器(VCO) + 前置除頻器 (Prescaler) 輸出結果

TSPC Divider



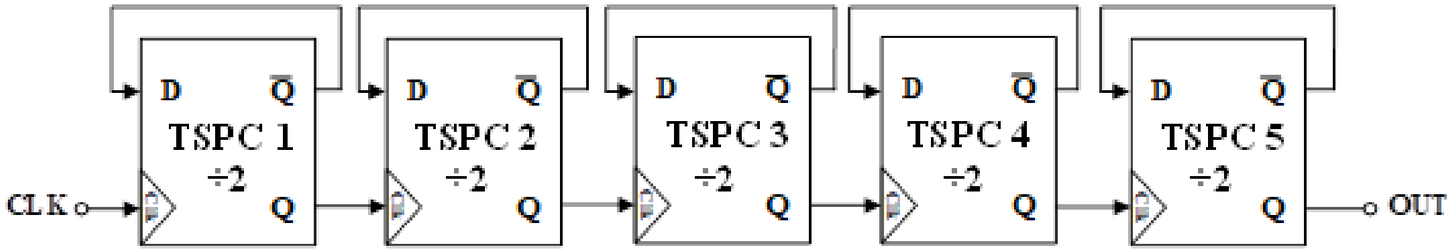


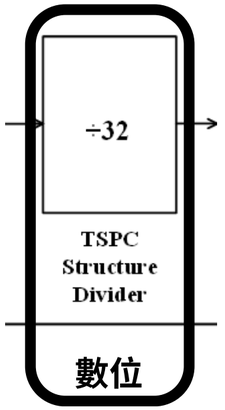


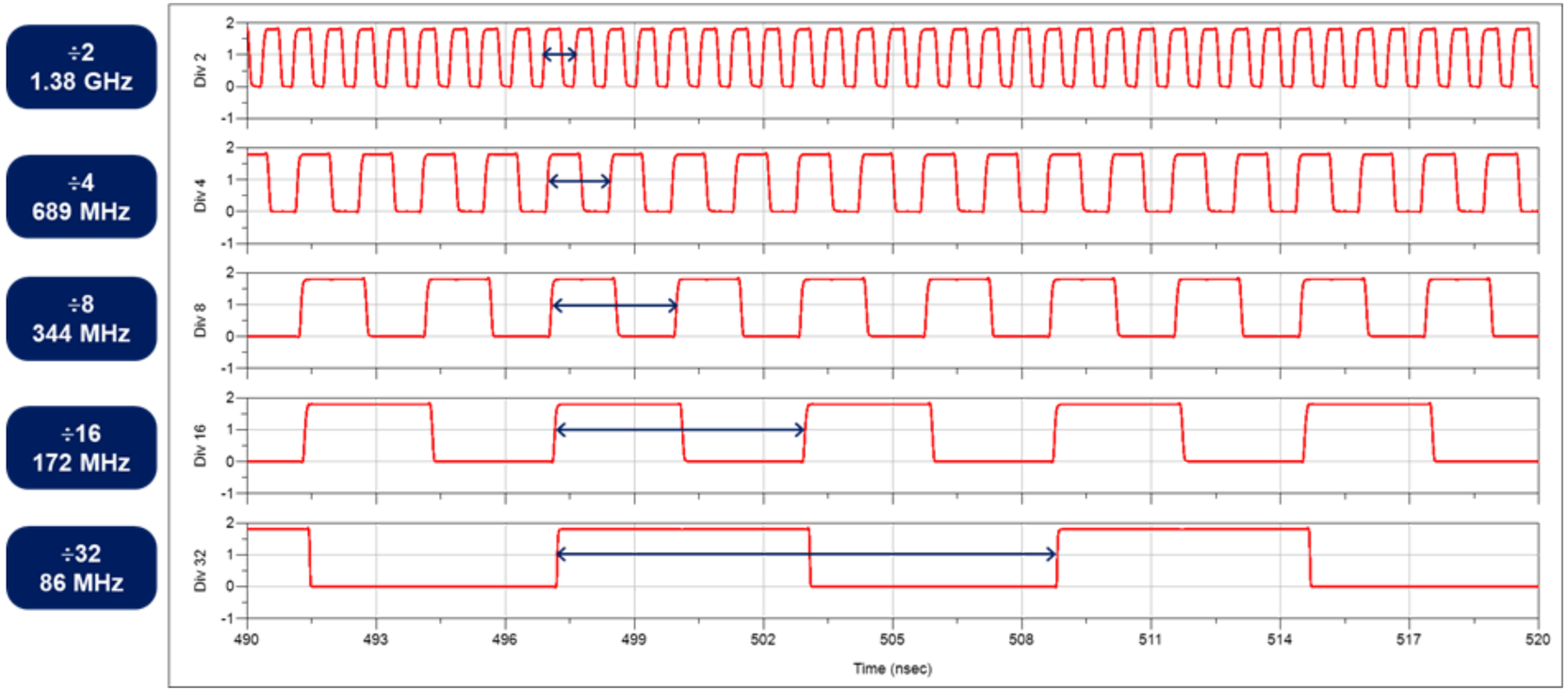
x5 顆

▲ 真單向時脈式除頻器 (TSPC Divider)

TSPC Divider 輸出

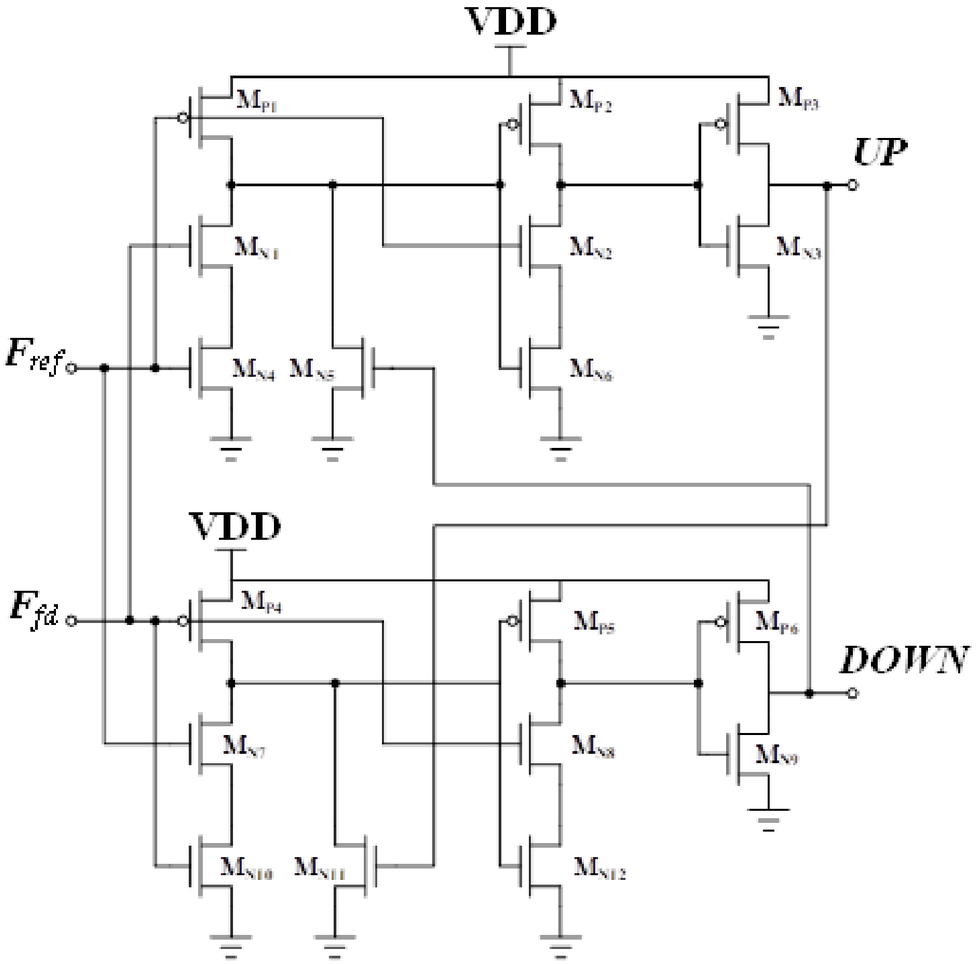


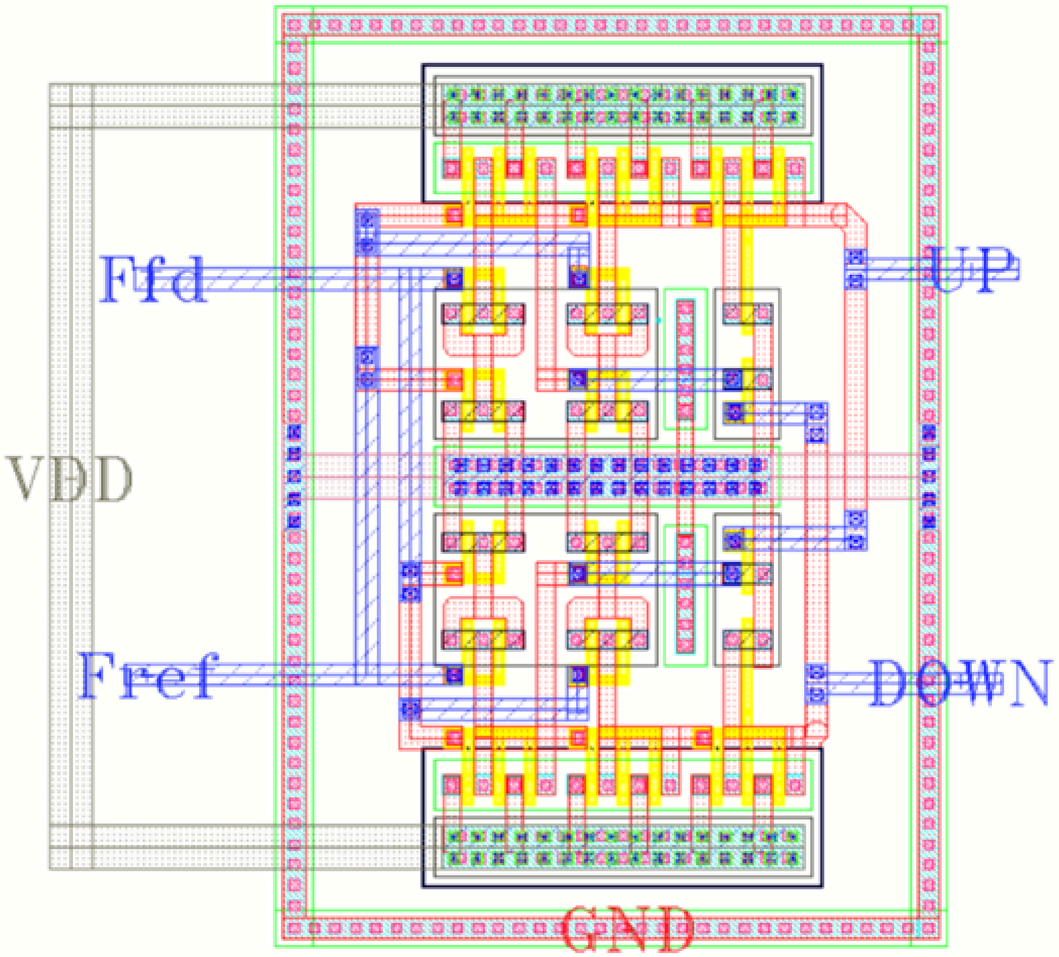




▲ 真單向時脈式除頻器 (TSPC Divider)

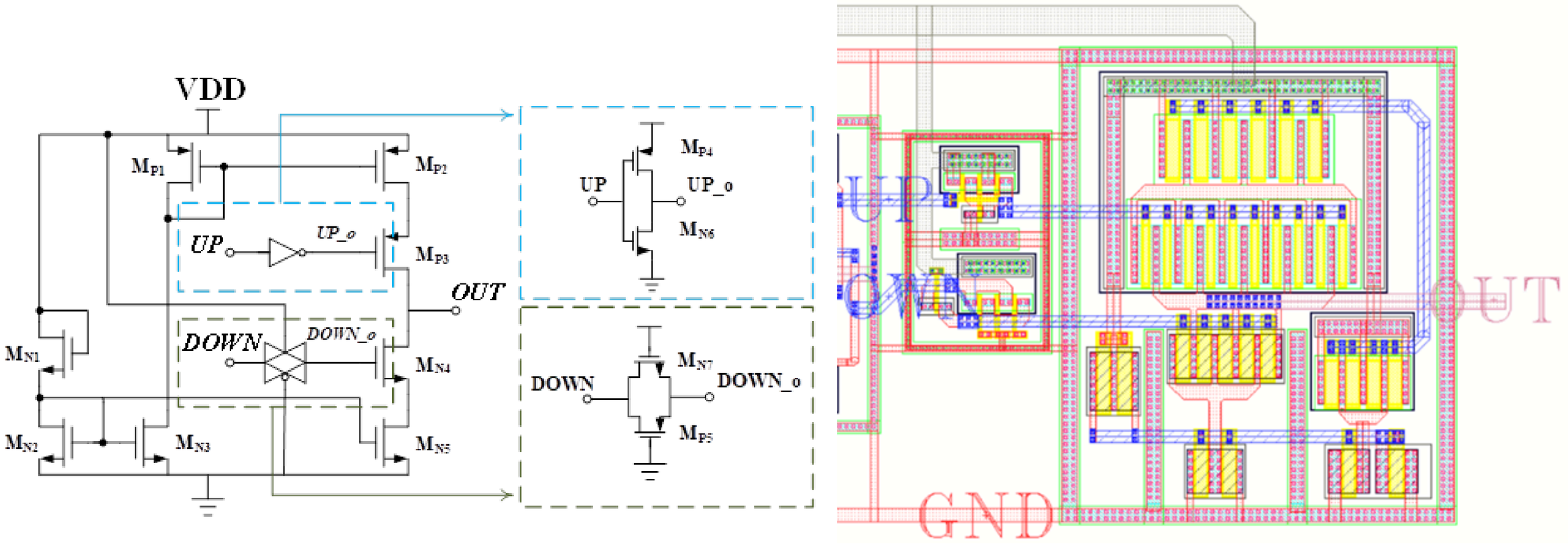
Layout 佈局





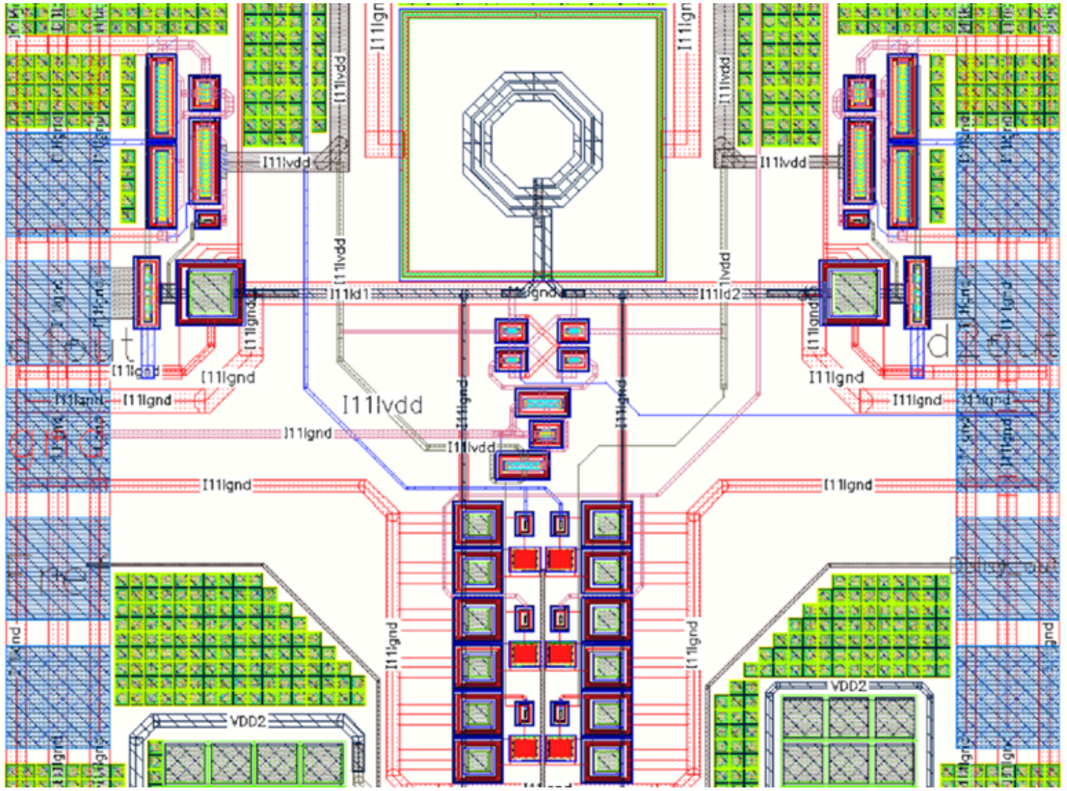
▲ 相位偵測器 (PFD)

Layout 佈局

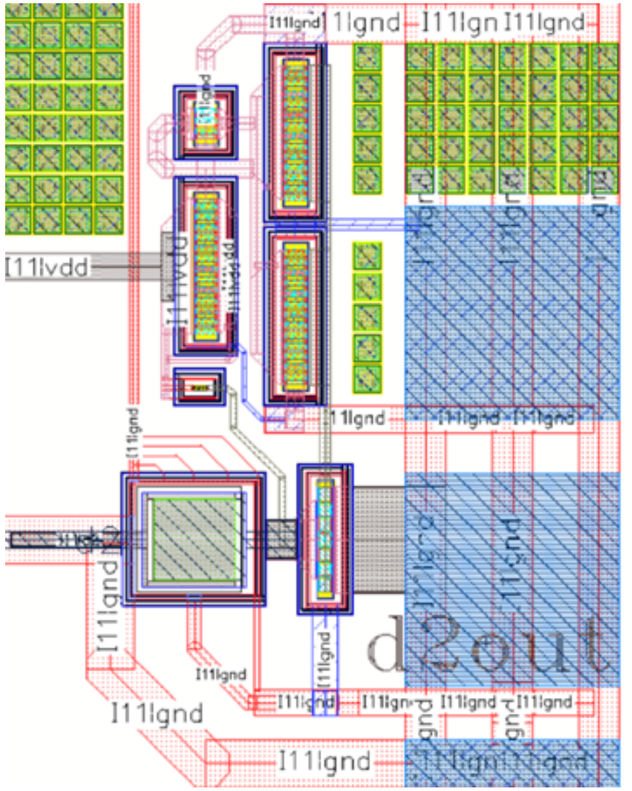


▲ 充電幫補 (CP)

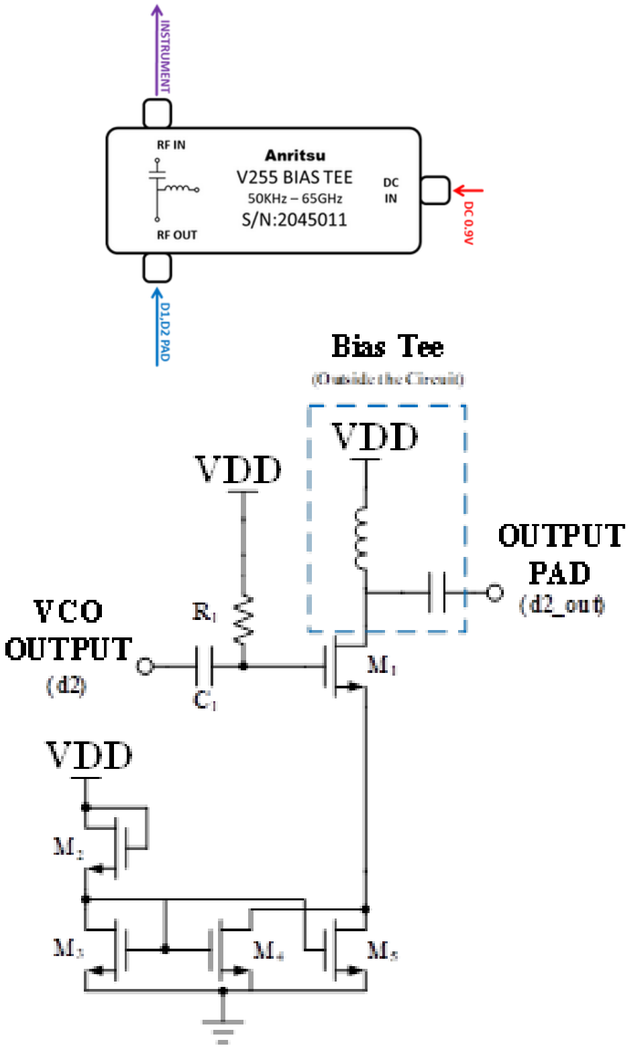
Layout 佈局



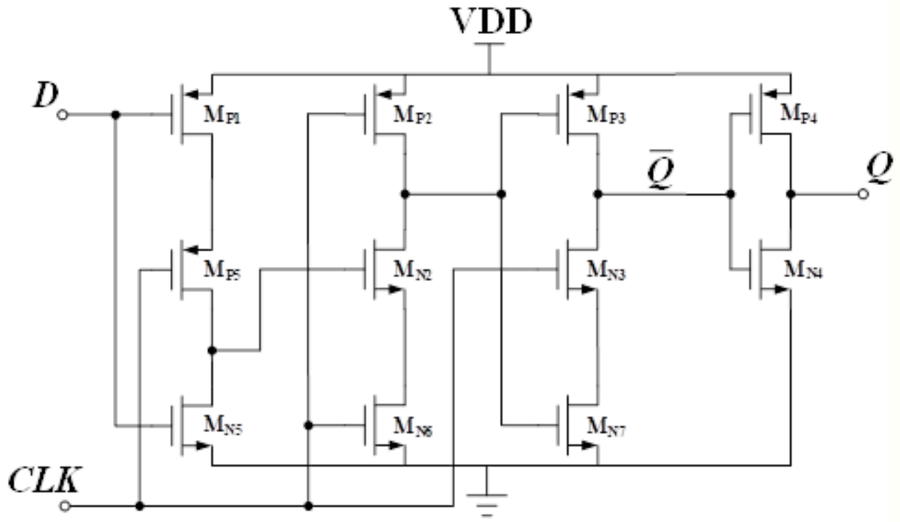
▲ 壓控振盪器 (VCO)

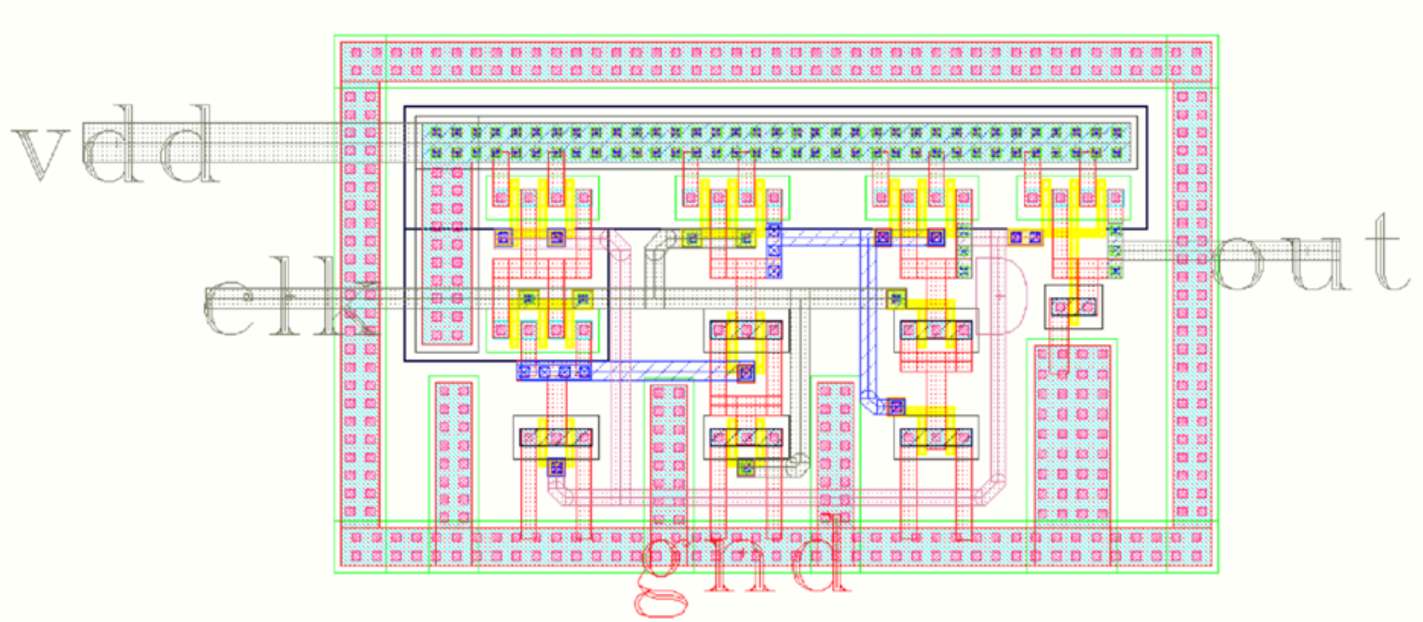


▲ 緩衝 (Buffer)



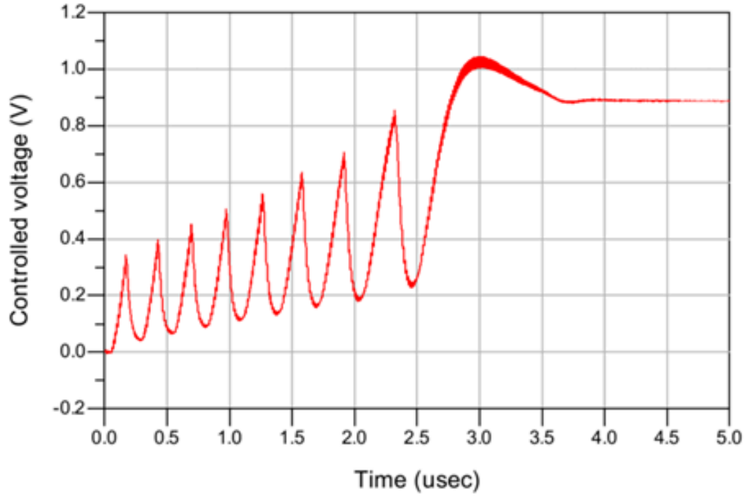
Layout 佈局

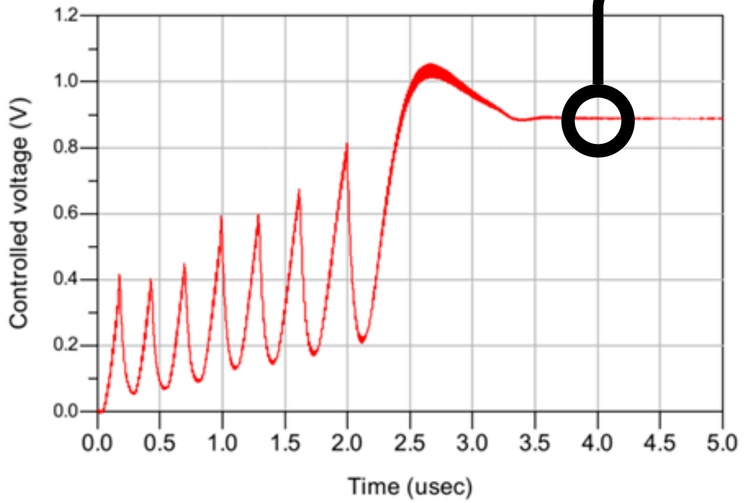




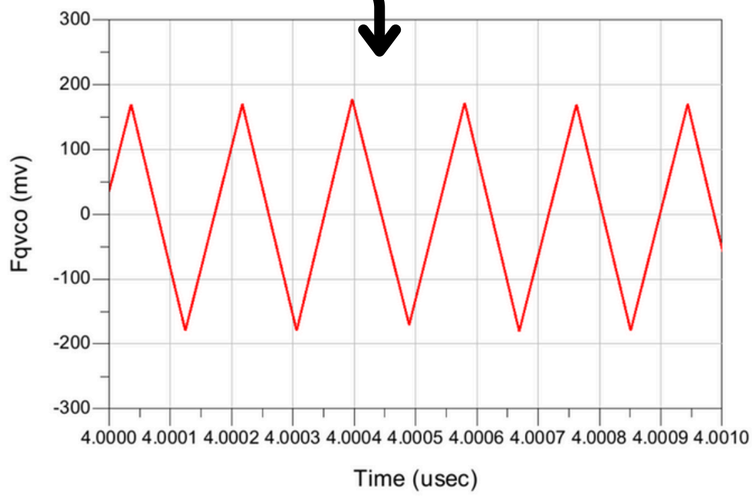
▲ 五級TSPC除頻器 (TSPC Divider)

PLL Post Layout Simulation

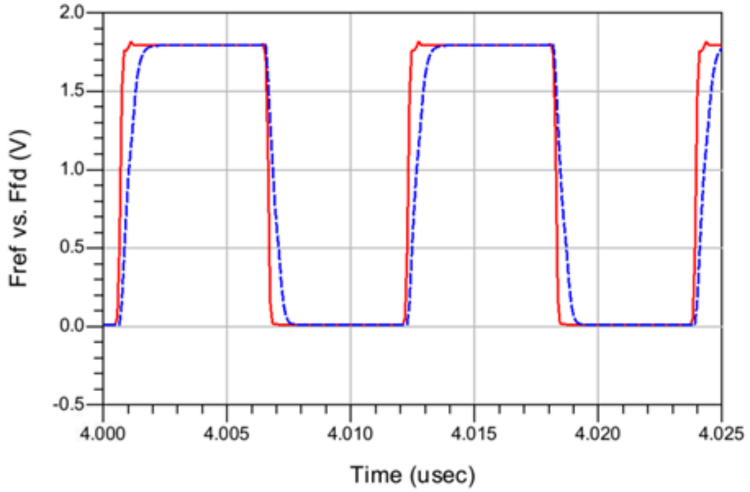


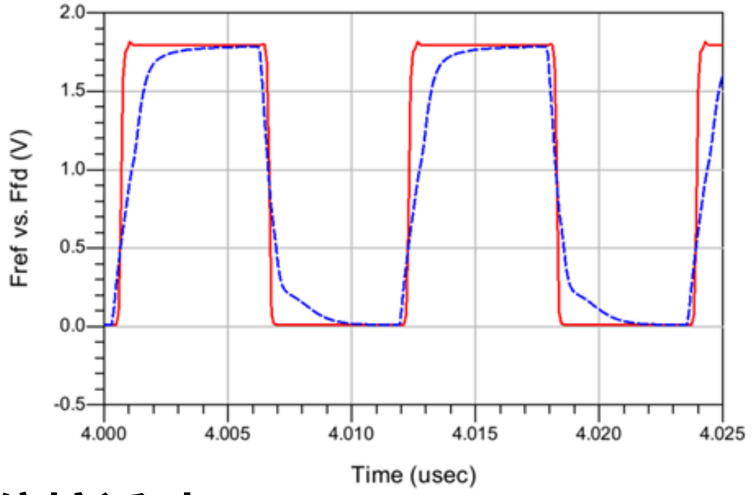


▲ 壓控振盪器控制電壓 (VCO Vctrl)

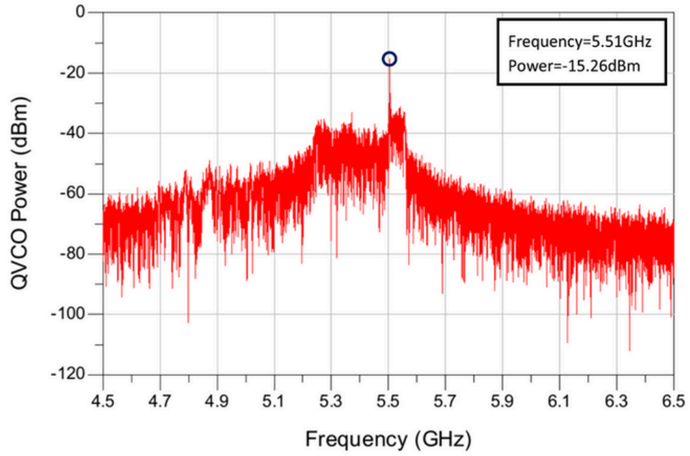


▲ VCO Vctrl 呈穩定態





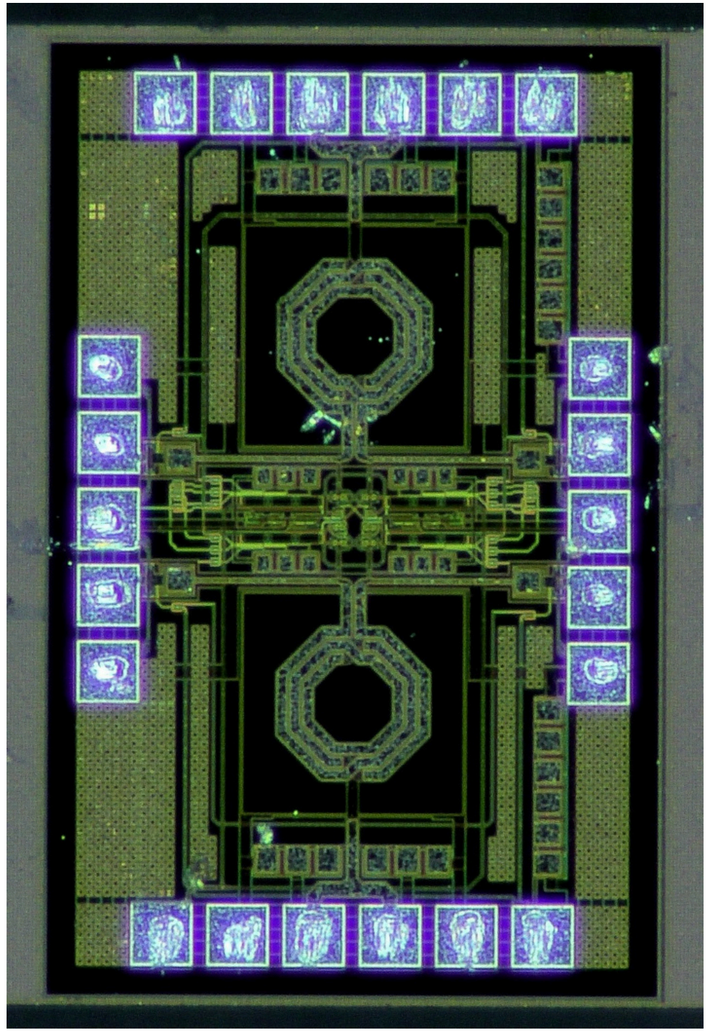
▲ 相位接近時



▲ VCO 頻率呈穩定態

Chip

0.72 mm*1.17 mm

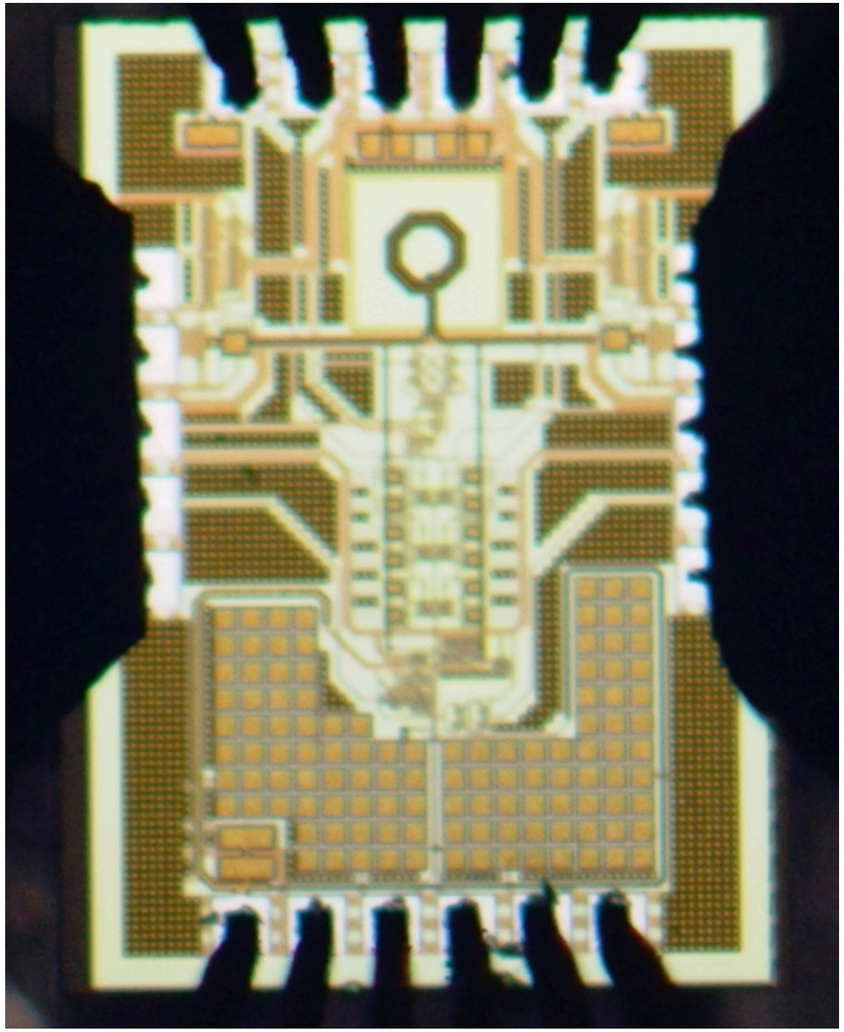


▲ 4相位壓控振盪器 (VCO)

Power:
Chip: 22.53 mW
57.6(buffer)mW

Partial work

0.82 mm*1.2 mm

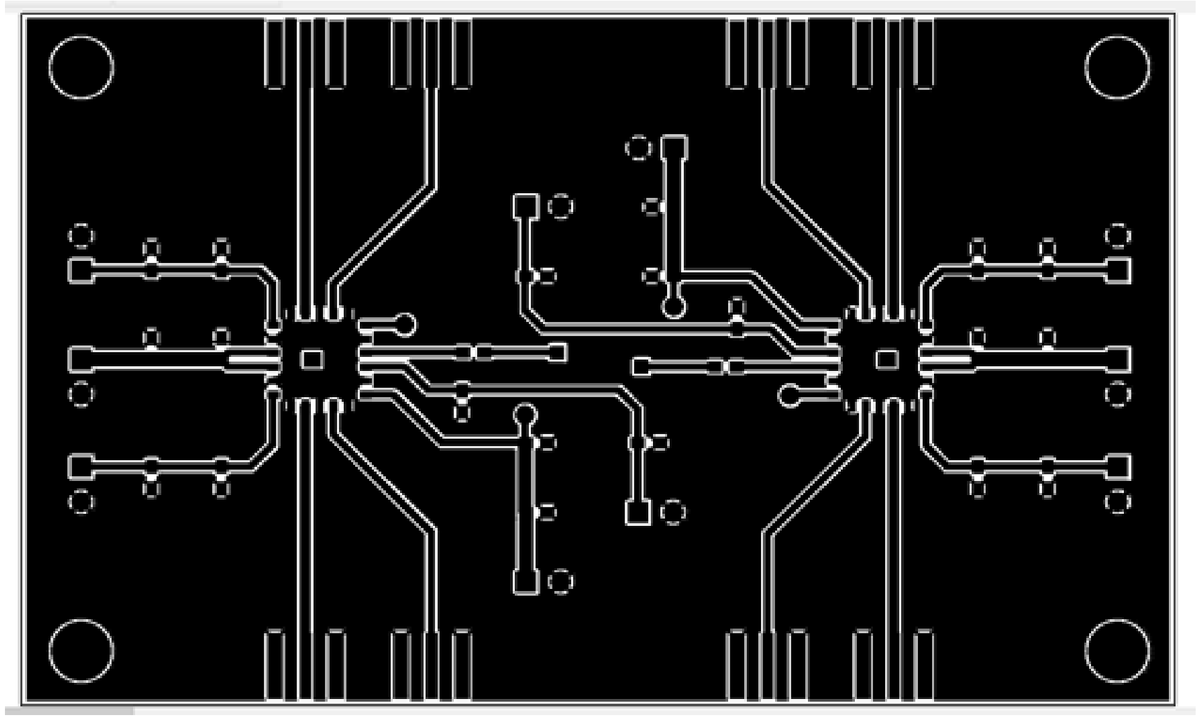


▲ 鎖相迴路 (PLL)

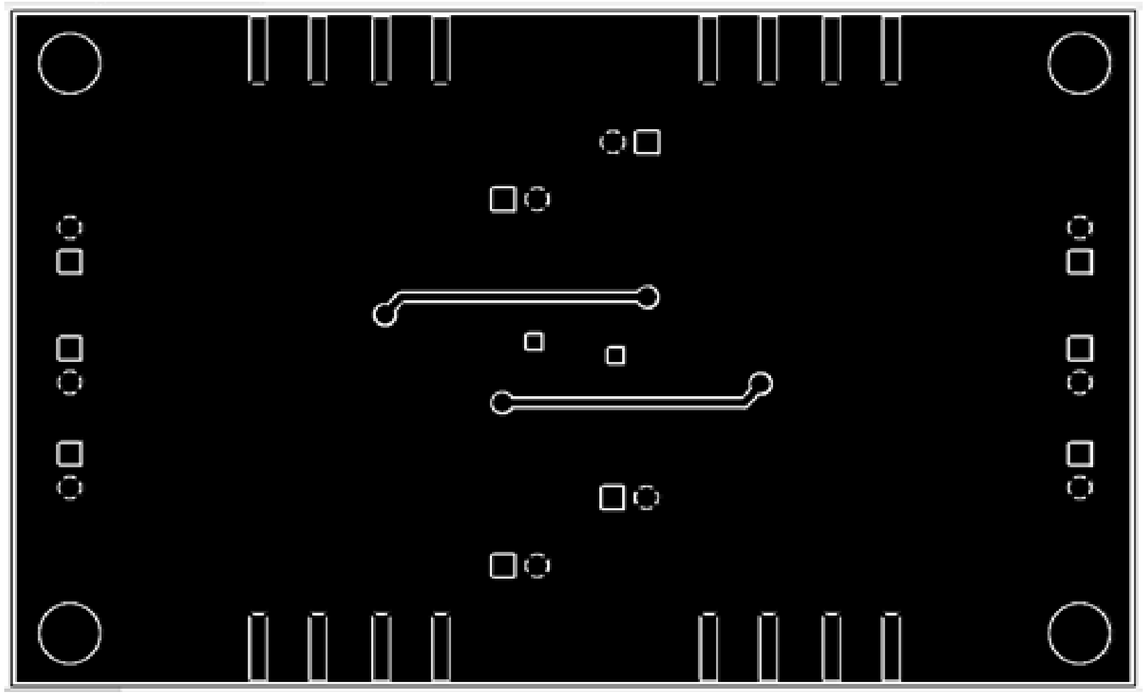
Power:
Chip: 15 mW
68(buffer)mW

Partial work

PCB Design

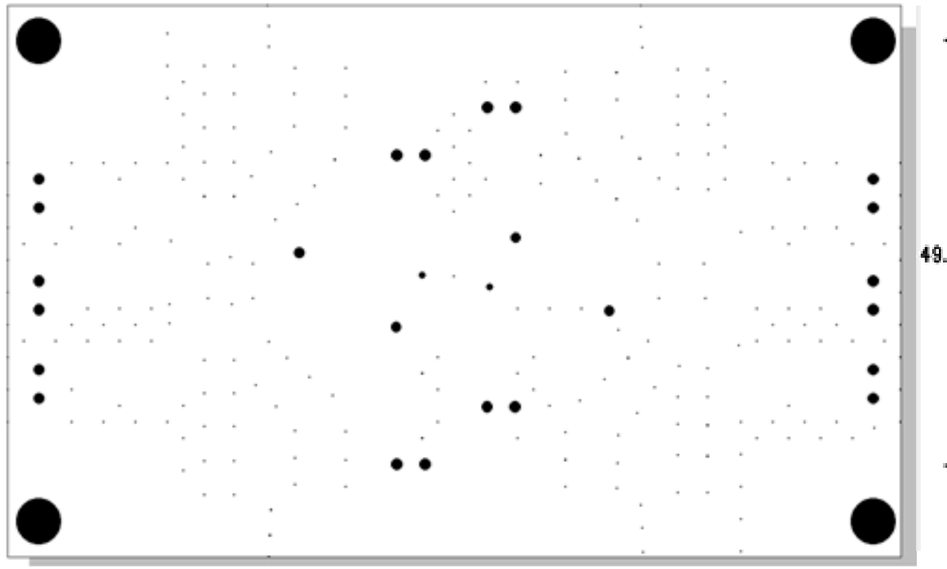


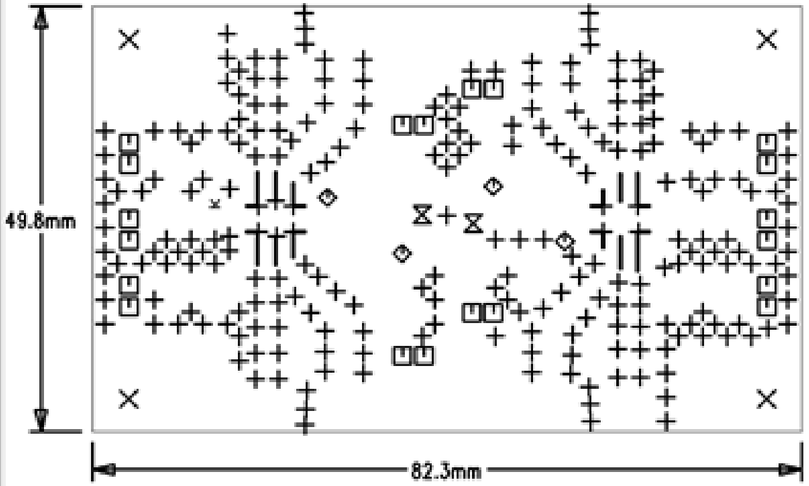
▲ PCB (TOP)



▲ PCB (BOTTOM)

PCB Design



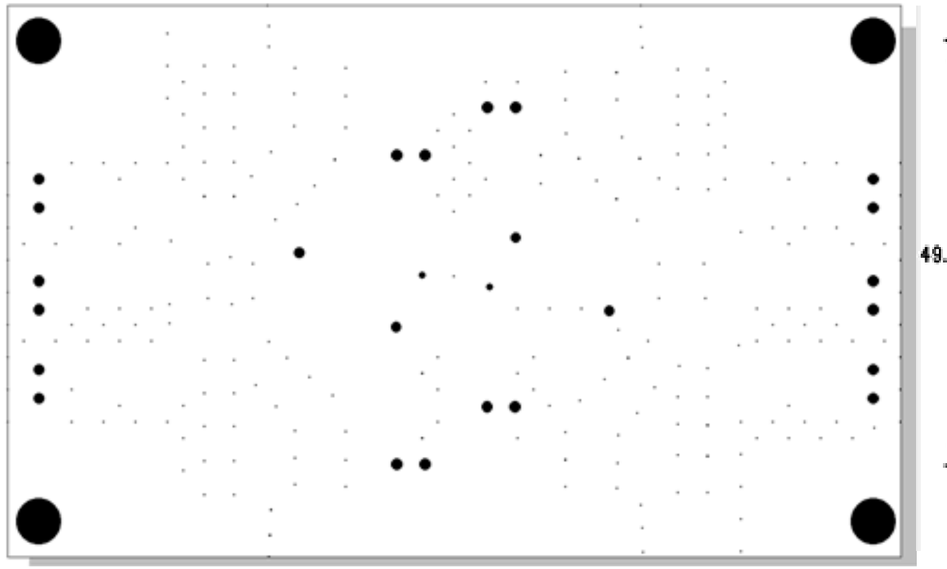


SIZE	QTY	SYM	PLATED	TOL
0.2	238	+	YES	+/-0.0
4	4	X	NO	+/-0.0
1	20	□	YES	+/-0.0
0.9398	4	◇	YES	+/-0.0
0.65	2	⧖	YES	+/-0.0

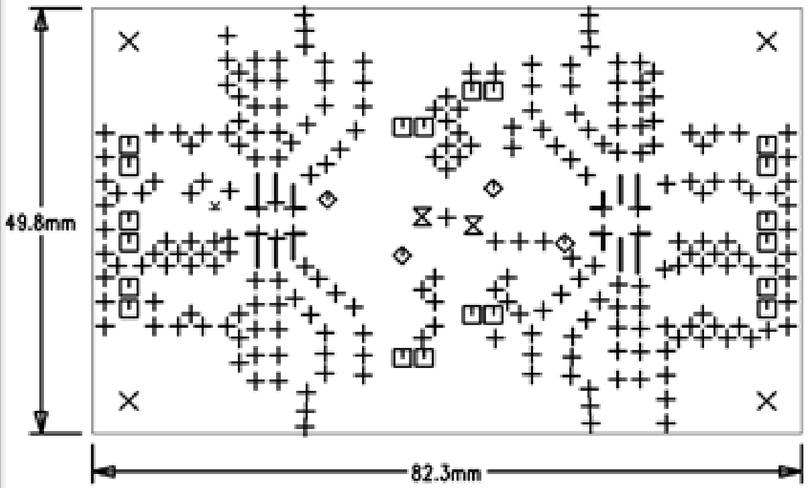
▲ PCB (NCD)

▲ PCB (DW)

PCB Design



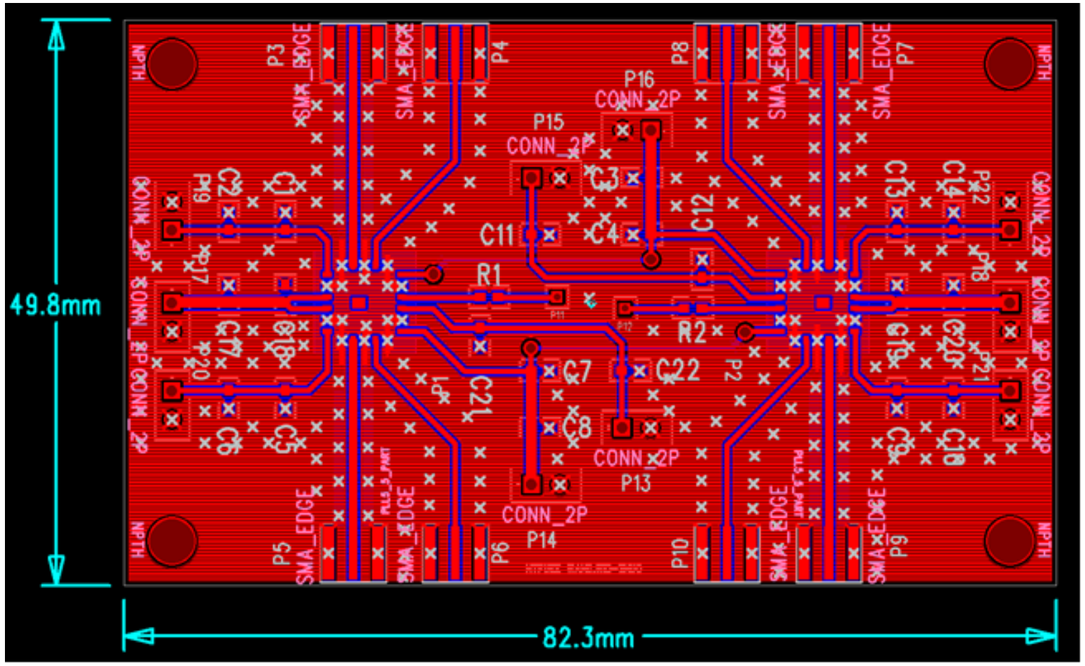
▲ PCB (NCD)



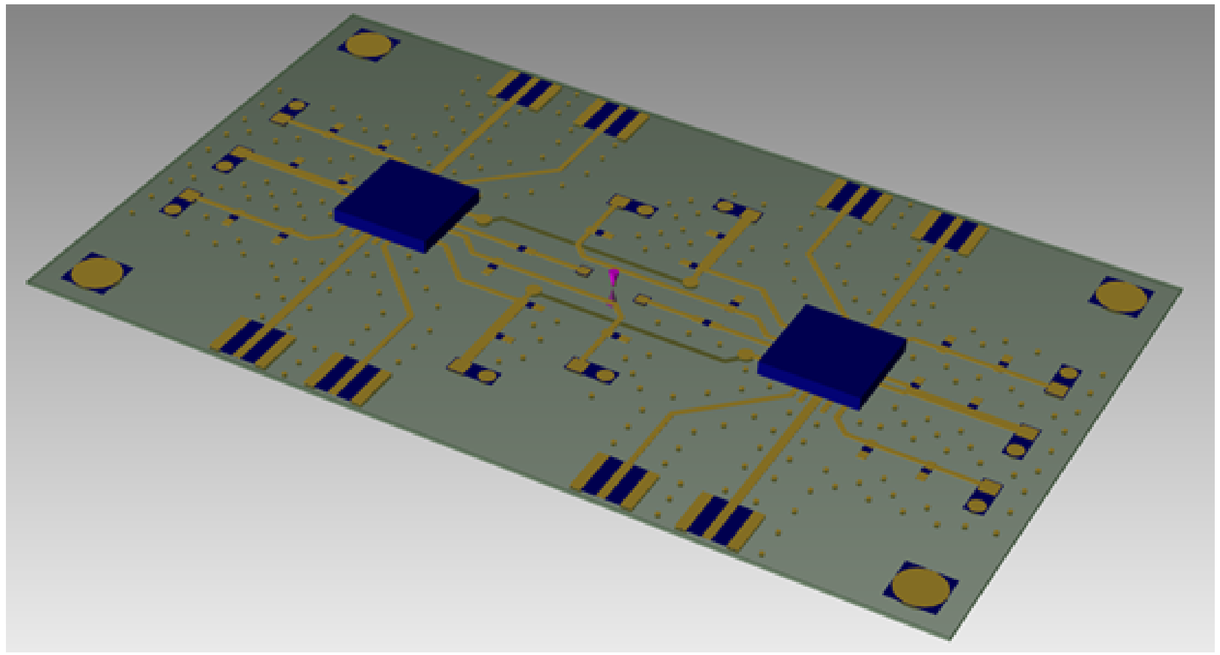
SIZE	QTY	SYM	PLATED	TOL
0.2	238	+	YES	+/-0.0
4	4	X	NO	+/-0.0
1	20	□	YES	+/-0.0
0.9398	4	◇	YES	+/-0.0
0.65	2	⧖	YES	+/-0.0

▲ PCB (DW)

PCB Design

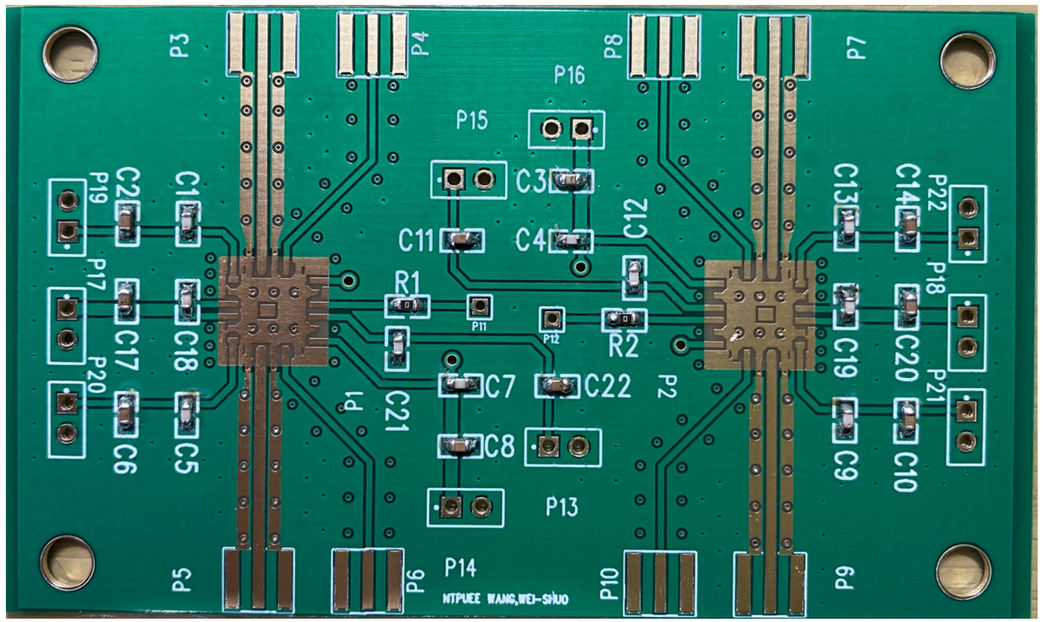


▲ PCB (2D)

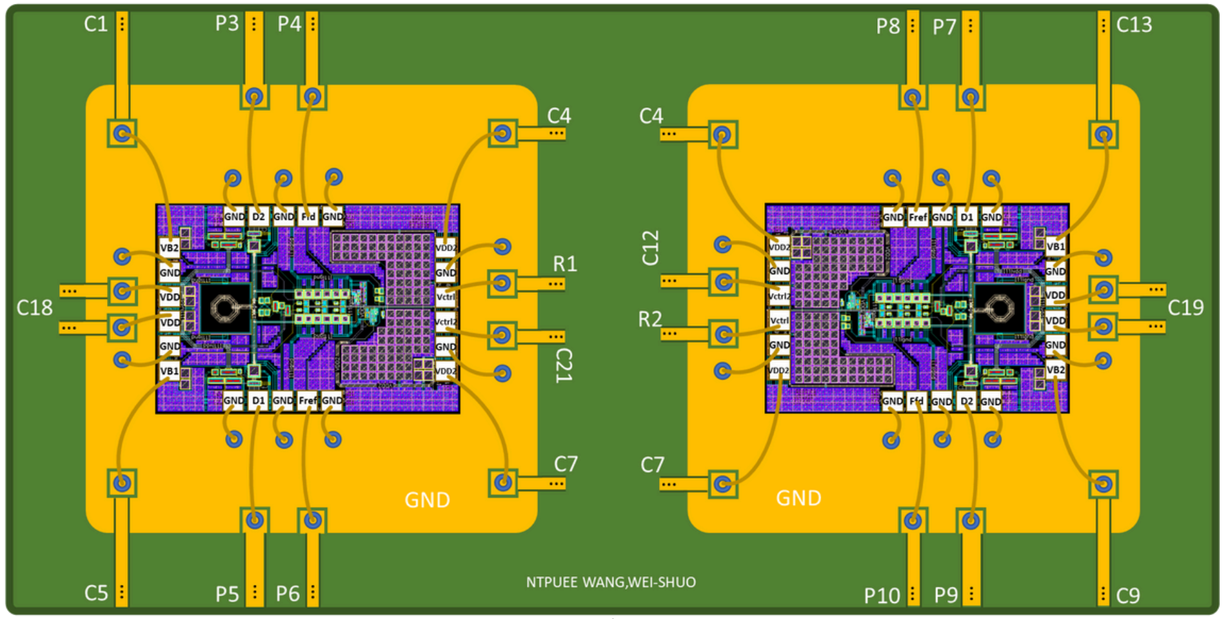


▲ PCB (3D)

PCB Bondwire

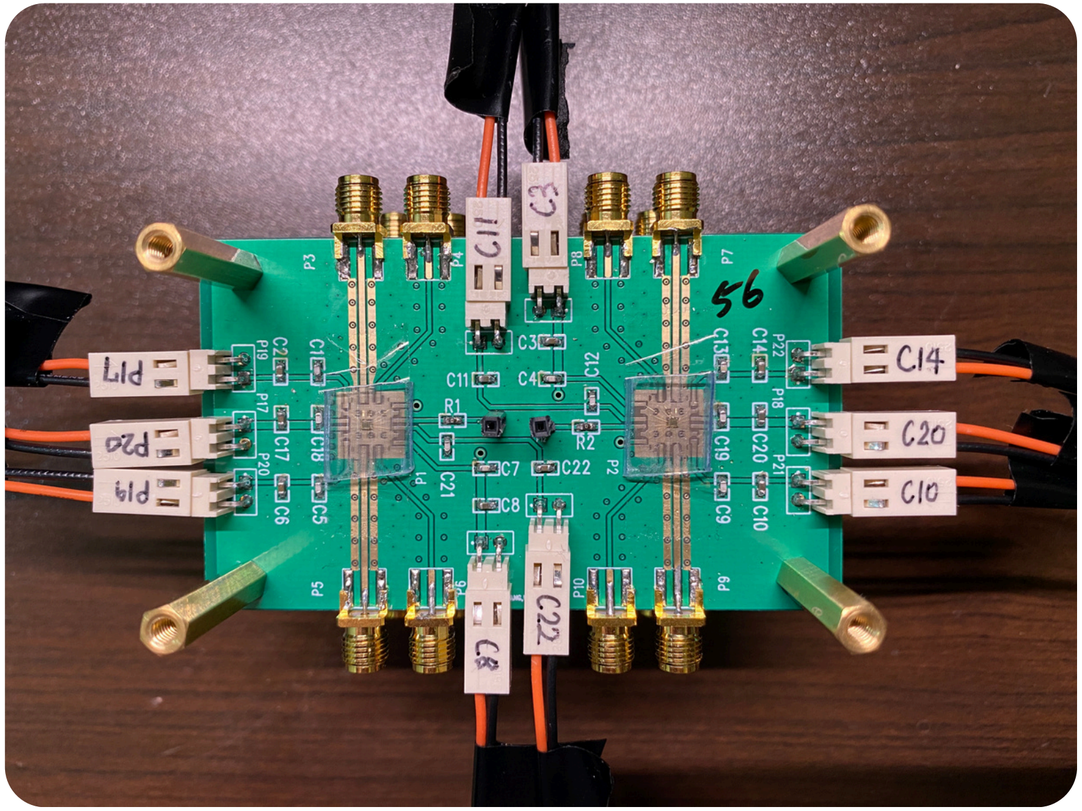


▲ PCB 製作完成

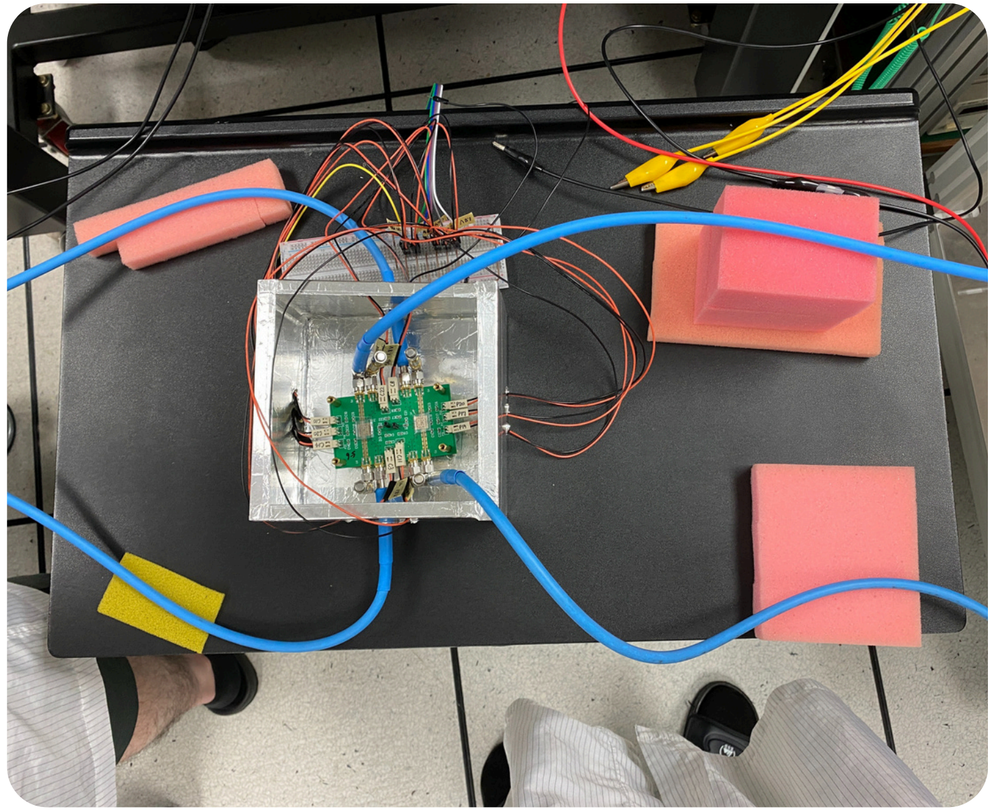


▲ 打線設計圖

PCB Bondwire



▲ 晶片流程全部完成



▲ 後續量測

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